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SURVEY

# Hybrid CNN-Transformer Models in Histopathology Image Analysis: A Scoping Review

**DOMINGOS LUCAS LATORRE DE OLIVEIRA[1,4], THAÍNA APARECIDA AZEVEDO TOSTA[2], LEANDRO ALVES NEVES[3], AND MARCELO ZANCHETTA DO NASCIMENTO[4]**

[1]Federal Institute of Education, Science and Technology of São Paulo (IFSP), São Paulo 01109-010, Brazil
[2]Science and Technology Institute, Federal University of São Paulo (UNIFESP), São José dos Campos 12247-0142, Brazil
[3]Department of Computer Science and Statistics (DCCE), São Paulo State University (UNESP), São José do Rio Preto 15054-000, Brazil
[4]Faculty of Computer Science (FACOM), Federal University of Uberlândia (UFU), Uberlândia 38400-902, Brazil

Corresponding author: Domingos Lucas Latorre de Oliveira (domingos.oliveira@ifsp.edu.br)

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**ABSTRACT** Hybrid convolutional neural networks (CNNs) and Transformer-based architectures have demonstrated strong potential in histopathological image analysis by combining local feature extraction with global context modeling. This dual capability is critical in computational pathology, where accurate interpretation depends on both fine-grained morphological details and broad tissue-level patterns. To systematically assess how these hybrid models have been applied, this scoping review maps the current landscape of CNN-Transformer architectures across classification, segmentation, and detection tasks in histopathological images. Following the PRISMA-ScR guidelines and the Joanna Briggs Institute methodology, a comprehensive literature search was conducted across Web of Science, IEEE Xplore, Scopus, and PubMed. A total of 39 peer-reviewed journal articles published from 2020 onward met the inclusion criteria. To guide the analysis, a task-specific taxonomy was proposed to classify hybrid architectures based on their primary application, offering a structured framework for model comparison and evaluation. Thus, the review synthesizes key aspects of model implementation, including architectural designs, training strategies, dataset characteristics, and evaluation protocols. Findings reveal that hybrid models frequently outperform conventional approaches; however, critical limitations persist, particularly regarding external validation, model interpretability, and reproducibility. Notably, only a small subset of studies employed rigorous validation schemes or integrated explainability into clinical workflows. Moreover, by consolidating evidence and exposing methodological gaps, this review contributes a comprehensive foundation for advancing hybrid deep learning models in digital pathology. It offers actionable insights to inform the development of scalable, transparent, and clinically aligned computational tools for histopathological image analysis.

**INDEX TERMS** Hybrid CNN-transformer, vision transformer, deep learning, histopathology image analysis, medical image processing, scoping review.

## I. INTRODUCTION

Histopathological analysis is essential in contemporary medical diagnostics, particularly for cancer detection, infectious disease identification, and therapeutic planning [1], [2]. In oncology, histological evaluation often constitutes the primary basis for defining diagnosis, prognosis, and treatment strategies, underscoring its central role in guiding patient management [1].

The global burden of disease has intensified the demand for accurate, scalable, and efficient diagnostic tools. In 2022 alone, there were an estimated 20 million new cancer cases and 9.7 million related deaths [3]. Breast cancer led global incidence, followed by prostate and oral cavity

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cancers, with the latter disproportionately affecting South and Southeast Asian populations [4], [5]. These figures underscore the growing workload faced by pathology services and the critical need for automated, high-throughput image analysis systems.

Traditionally, tissue evaluation is conducted by pathologists through visual inspection of hematoxylin and eosin (H&E)-stained slides under a microscope [6]. The advent of digital slide scanners has transformed this workflow, enabling the conversion of specimens into high-resolution whole-slide images (WSIs) that preserve both cellular detail and global tissue architecture. WSIs allow detailed assessment of nuclear morphology, spatial tissue organization, and cell distribution, key features in diagnostic and prognostic interpretation [7].

This digitization has opened new opportunities for computational pathology. While classical computer vision approaches laid the groundwork, the widespread availability of WSIs has fueled the development of advanced machine learning algorithms tailored to histopathological tasks [8], [9]. These models now form the backbone of computer-aided diagnosis (CAD) systems, supporting tasks such as disease detection, tissue segmentation, and tumor grading [10], [11]. By providing quantitative outputs and highlighting diagnostically relevant regions, CAD systems assist pathologists in high-volume settings, improving consistency and reducing inter-observer variability [12], [13].

In this context, deep learning (DL), especially convolutional neural networks (CNNs), has become foundational in CAD pipelines for histopathology [14]. CNNs exploit local connectivity and weight sharing to detect fine-grained morphological features such as nuclei, glands, and stromal components [15]. However, their inductive bias toward locality constrains their ability to model long-range dependencies and global tissue architecture, which are features often critical for accurate interpretation in whole-slide image analysis [16].

To address this limitation, Vision Transformers (ViTs) were introduced, extending the self-attention mechanism to image analysis [17], [18]. By modeling pairwise relationships across entire images, ViTs offer global contextualization but lack built-in spatial priors and require large annotated datasets to achieve reliable performance, conditions that are often unmet in medical imaging domains [19].

Hybrid CNN-Transformer architectures have been proposed as a promising solution, combining the locality-aware inductive biases of CNNs with the global modeling capabilities of Transformers [20], [21], [22]. In these models, convolutional blocks typically extract localized features, while the attention mechanisms integrate global context. Various integration strategies (sequential, parallel, integrated, and cross-guided) have demonstrated effectiveness in key histopathological tasks, including classification, segmentation, and detection [23], [24], [25].

Despite their growing relevance, existing literature reviews often focus on CNN-based approaches [15], [26], pure Transformer models [19], [27], or broadly discuss deep learning in medical imaging [11], [28]. While a few recent surveys explore hybrid architectures [24], [28], they typically lack specificity in histopathological applications. To the best of our knowledge, no prior work has systematically mapped hybrid CNN-Transformer models in digital pathology. Table 1 outlines the scope and limitations of relevant reviews, underscoring the need addressed by the present study.

Given the architectural diversity and rapid evolution of hybrid designs, a scoping review is well suited to catalog current approaches, classify applications, and identify methodological gaps in this developing field [32]. The objective of this review is to systematically map the literature on hybrid CNN-Transformer models applied to histopathology image analysis. Specifically, we aim to:

i. Categorize the types of hybrid CNN-Transformer architectures proposed for histopathological image processing;
ii. Analyze their use in core tasks such as classification, segmentation, and detection;
iii. Identify methodological limitations and research gaps to inform future directions in the domain.

By mapping the current state of research, the central question guiding this review is: *How have hybrid CNN-Transformer models been applied to histopathology image analysis, and what are their key benefits and limitations across classification, segmentation, and detection tasks?* Based on this inquiry, we formulated five specific research questions:

- **RQ1**: Which hybrid CNN-Transformer architectures have been proposed for histopathology image analysis, and how do they integrate convolutional and attention components?
- **RQ2**: What datasets are employed to train and evaluate these models across different tasks?
- **RQ3**: Which evaluation metrics are most commonly reported?
- **RQ4**: What advantages and limitations, e.g., in terms of computational cost, generalization, and scalability, are associated with hybrid models?
- **RQ5**: What trends and open challenges are shaping future research in this area?

In this context, the main contributions of this scoping review are:

- A task-based taxonomy that organizes hybrid CNN-Transformer architectures by their primary application in histopathological image analysis (classification, segmentation, or detection).
- A comprehensive synthesis of implementation practices, covering architectural configurations, training protocols, datasets, and evaluation metrics, with an emphasis on reproducibility and methodological rigor.
- The identification of key limitations across the literature, such as over-reliance on internal validation, limited adoption of interpretability frameworks, and lack of

**TABLE 1.** Prior reviews covering CNNs, Transformers, and hybrid models in histopathology.

| Year | Study | Period Covered | Primary Focus | Gap Relative to This Review |
|---|---|---|---|---|
| 2021 | Srinidhi et al. [15] | 2015–2021 | Deep learning in histopathology | Emphasizes CNNs; omits Transformers and hybrids |
| 2022 | Banerji and Mitra [26] | 2016–2022 | CNN applications in histopathology | Omits ViTs and hybrid models |
| 2022 | Abdelsamea et al. [29] | 2015–2022 | AI in histopathology | Broad AI survey; hybrids not discussed in depth |
| 2022 | Kim et al. [30] | 2000–2022 | AI trends in pathology | Broad perspective; no hybrid CNN–Transformer analysis |
| 2023 | Atabansi et al. [27] | 2019–2023 | Transformers in histopathology | Focuses on pure ViTs; excludes hybrids |
| 2023 | Springenberg et al. [31] | 2016–2023 | CNNs vs. ViTs in histopathology | Compares CNNs and ViTs; hybrids not covered |
| 2023 | Khan et al. [28] | 2017–2023 | ViTs and hybrid CNN–Transformers | General survey; not histopathology-specific |
| 2023 | Xu et al. [19] | 2021–2023 | Vision Transformers in histopathology | Pure ViTs; no hybrid coverage |
| 2024 | Xu et al. [11] | 2020–2024 | Medical image segmentation methods | Includes hybrids, but not focused on histopathology |
| 2025 | Kim et al. [24] | 2020–2025 | Hybrids in radiology imaging | Radiology-centered; excludes histopathology |

benchmarking standards that constrain clinical translation.

- A structured overview of recent trends and unresolved challenges, providing practical insights for the development and evaluation of future hybrid models in digital pathology.

## II. BACKGROUND

This section provides the fundamental context for our scoping review of hybrid CNN-Transformer models in computational histopathology.

### A. HISTOPATHOLOGY IMAGE ANALYSIS

To visualize cellular and tissue architectures, which are typically translucent, biological tissues require staining. The most prevalent method is H&E staining, where Hematoxylin imparts a purplish-blue color to cell nuclei, and Eosin stains cytoplasm and the extracellular matrix (stroma) in varying shades of pink [6]. This differential staining is crucial for the morphological assessment of tissue. In specific contexts, Immunohistochemistry (IHC) is another important technique. IHC employs antibodies to detect specific proteins or biomarkers within tissue sections, often yielding critical diagnostic, prognostic, or predictive insights and supporting tasks like biomarker expression quantification [33].

A key development in histopathology image analysis is the advent of WSIs, which are high-resolution digital captures of entire glass slides produced by specialized scanners [7]. These images are notable for their massive data footprint, often reaching gigapixel scales, and their inherent multi-resolution (pyramidal) structure that stores the image at various magnification levels. The sheer size and multi-magnification nature of WSIs present substantial computational hurdles, making it generally impractical for deep learning models to process an entire WSI at its native, highest resolution in a single pass. Consequently, analysis is frequently conducted on patches or tiles, smaller and more manageable image regions extracted from the WSI, usually at a specific magnification, which serve as direct inputs for computational models (see Fig. 1). This modular, patch-based approach is well-aligned with deep learning methods, particularly CNNs and ViTs, which process spatially structured data. The next subsections describe these architectures in detail and explore their respective strengths and limitations when applied to histopathology images.

#### 1) COMMON COMPUTATIONAL TASKS

Computational models, particularly those based on DL, are widely applied to histopathology images to perform key analysis tasks such as segmentation, classification, and detection (see Fig. 2).

Segmentation focuses on identifying and precisely delineating the boundaries of specific regions of interest. This includes the segmentation of individual cell nuclei, vital for morphometric analysis, cell counting, and the assessment of nuclear atypia; the outlining of glandular structures, particularly important in adenocarcinomas like those of the colon or prostate; and the differentiation of broader tissue regions, such as tumor areas, stroma, and normal tissue, as well as other structures like blood vessels or areas of necrosis. Segmentation tasks can be further divided into semantic segmentation (labeling each pixel with a class) and instance segmentation (distinguishing individual object instances within the same class) (see Fig. 2b). Mathematically, given an image $I$ composed of pixels $p$, segmentation aims to assign a class label $c_j$ from a set of classes $C = \{c_1, c_2, \ldots, c_m\}$ to each pixel $p_i \in I$. The output is a segmentation map $S$ of the same dimensions as $I$, where each element $S_{xy}$ contains the class label for the pixel at coordinates $(x, y)$. Thus, $S_{xy} \in C$ [34], [35].

Classification involves assigning a descriptive label to an image or a specific region [23], [36], [37]. This can occur at the patch-level, for instance, by categorizing individual image patches as tumorous versus non-tumorous or benign versus malignant (see Fig. 2c). At a broader scope, WSI-level classification aims to provide a diagnosis for the entire

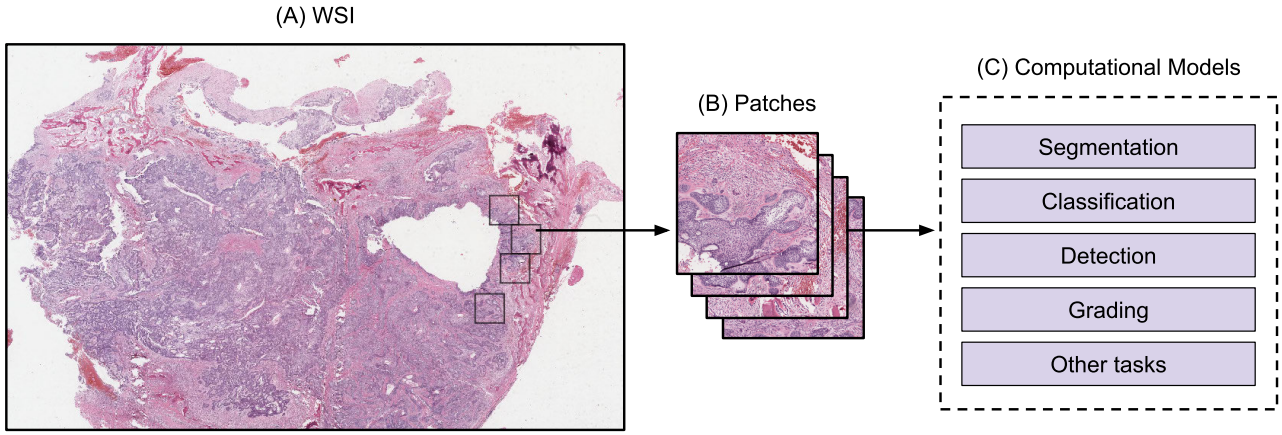
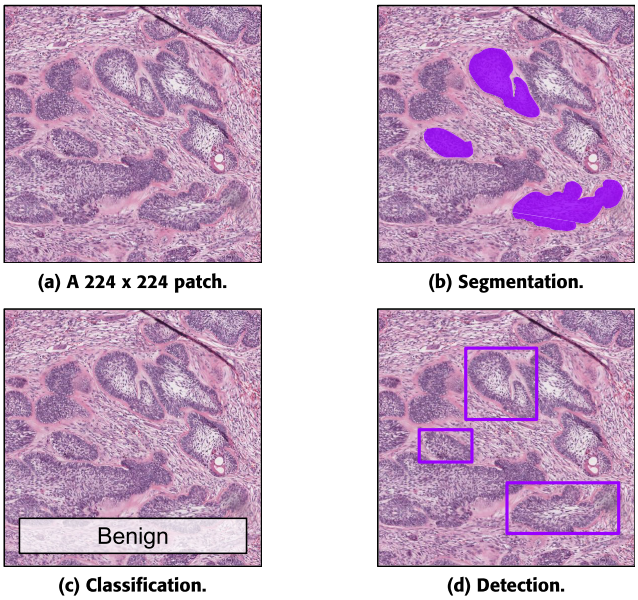



**FIGURE 1.** Typical computational-pathology workflow. (A) Acquisition of a WSI. (B) Extraction of regions of interest as patches. (C) Processing by algorithms for segmentation, classification, detection, grading and related tasks.




(a) A 224 x 224 patch. (b) Segmentation. (c) Classification. (d) Detection.

**FIGURE 2.** Common image analysis tasks in computational pathology.

slide, such as determining the presence or absence of cancer, performing tumor grading (e.g., Gleason scoring for prostate cancer), or predicting molecular subtypes, biomarker status, and even patient outcomes based on learned morphological features. In addition to single-label classification, many applications involve multilabel classification, where multiple relevant attributes (e.g., tumor type, grade, and mutation status) are predicted simultaneously. Mathematically, if $I$ represents an input image (or patch) and $Y = \{y_1, y_2, \ldots, y_k\}$ is a finite set of $k$ predefined class labels, the classification task is to learn a function $f : I \rightarrow Y$ that maps the input $I$ to its correct label $y \in Y$. For a given image $I_i$, the model predicts a label $\hat{y}_i = f(I_i)$.

Object detection is also frequently employed to localize specific instances of objects within an image (see Fig. 2d) [35]. Common examples include the detection of mitotic figures, which are cells undergoing mitosis and serve as a key indicator for tumor proliferation and grading [38]; the identification of tumor-infiltrating lymphocytes (TILs) within or around tumor tissue, which can hold prognostic and therapeutic significance [39]; and the detection of specific cells expressing particular biomarkers, often highlighted by IHC [40]. Mathematically, for an input image $I$, object detection aims to produce a set of $N$ tuples $O = \{(B_j, c_j)\}_{j=1}^{N}$, where each $B_j$ is a bounding box (e.g., defined by its top-left corner coordinates $(x, y)$, width $w$, and height $h$) that localizes an object instance, and $c_j$ is the class label of the detected object from a predefined set of classes $C = \{c_1, c_2, \ldots, c_m\}$. The number of objects $N$ is not previously known.

While these tasks form the foundation of computational histopathology, their effective execution is often hindered by practical and technical challenges, which are discussed in the following subsection.

### 2) KEY CHALLENGES

The application of computational methods to histopathology images involves several inherent challenges that directly influence model design, training, and clinical adoption. A primary limitation arises from substantial variability in tissue preparation, staining protocols, and scanning instrumentation, which introduces heterogeneity in color distribution, intensity levels, and spatial resolution [41]. These inconsistencies hinder generalization, especially when models trained on a single dataset are applied to data from different institutions [42]. Moreover, artifacts such as tissue folds, tears, and blurred regions frequently compromise image quality, further complicating automated analysis [43].

Another critical aspect is the multi-scale nature of histopathological structures. Relevant diagnostic information may appear at different levels of magnification, from cellular-level features such as nuclear morphology and chromatin texture to glandular architecture and overall tissue organization [44], [45]. Consequently, effective models must be capable of extracting and integrating contextual information across multiple spatial resolutions, a requirement that challenges conventional convolutional designs and motivates hybrid approaches.

Model interpretability remains a key barrier to clinical integration. Despite achieving high predictive accuracy, many hybrid CNN-Transformer models operate as opaque systems, limiting trust among pathologists who require transparent, justifiable outputs to support diagnostic decisions [46]. To mitigate this limitation, techniques such as attention visualization, Grad-CAM, and SHAP have been proposed to identify which regions or features contributed to a model's prediction [47], [48], [49]. However, these methods vary widely in implementation and reliability, and few studies evaluate their interpretability in a clinically meaningful context, limiting their adoption in diagnostic workflows [50].

Data availability and annotation constraints further restrict model development. Building generalizable DL models requires access to large, diverse, and well-annotated datasets, which are difficult to obtain in histopathology due to the time-consuming nature of expert labeling, institutional privacy regulations, and lack of standardized formats [15]. While public datasets such as TCGA, Camelyon, and PANDA offer important benchmarks, they also expose significant inter-institutional variability in staining techniques, image resolution, and diagnostic criteria. This diversity presents a substantial obstacle to model robustness across clinical environments. As a result, normalization techniques, augmentation pipelines, and domain adaptation strategies are increasingly employed, although their adoption remains inconsistent across studies [51], [52].

In this context, these challenges collectively underscore the need for models that are not only accurate and efficient but also interpretable, robust to domain shifts, and validated across heterogeneous datasets. Current hybrid CNN-Transformer architectures have begun to address some of these demands, yet limitations in reproducibility, explainability, and cross-domain generalization continue to constrain their translational impact in clinical pathology.

### B. CONVOLUTIONAL NEURAL NETWORKS

CNNs are a class of DL models specifically designed to process data with a spatial or grid-like structure, such as images. Unlike fully connected neural networks that treat each input independently, CNNs leverage spatial hierarchies by using local receptive fields and weight sharing. Each convolutional layer applies a set of learnable filters that slide over the input to detect local patterns, such as edges and textures. These patterns are then composed hierarchically across successive layers to form increasingly abstract and semantically rich representations (see Fig. 3). Their inherent ability to learn hierarchical features from local pixel arrangements makes CNNs particularly well-suited for analyzing cellular morphology, texture, and other localized patterns within patches extracted from large WSIs.

In CNNs, both the input and the convolutional kernels are represented as tensors to support multiple channels. Let $\mathbf{I} \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ be an input image with $C_{\text{in}}$ channels, height $H$, and width $W$. A convolutional layer typically includes $C_{\text{out}}$ learnable kernels, each represented as $\mathbf{K}^{(c)} \in \mathbb{R}^{C_{\text{in}} \times k \times k}$, where $k$ is the kernel size and $c \in \{1, \ldots, C_{\text{out}}\}$. The output feature map $\mathbf{Y} \in \mathbb{R}^{C_{\text{out}} \times H' \times W'}$ is computed as:

$$\mathbf{Y}_c(i,j) = \sum_{d=1}^{C_{\text{in}}} \sum_{m=1}^{k} \sum_{n=1}^{k} \mathbf{K}_d^{(c)}(m,n) \cdot \mathbf{I}_d(i+m-1, j+n-1),$$

where $\mathbf{Y}_c(i,j)$ is the value at position $(i,j)$ of the $c$-th output channel, and $\mathbf{K}_d^{(c)}$ refers to the $d$-th input channel slice of the $c$-th kernel. A bias term $b_c$ is typically added, followed by a non-linear activation function such as the Rectified Linear Unit (ReLU), defined as $f(x) = \max(0, x)$ [35].

The main building blocks of CNNs include convolutional layers (which perform the convolution operation), nonlinear activation functions, and pooling layers (such as max-pooling or average pooling) that downsample the feature maps to reduce dimensionality and computational cost. These components are stacked in multiple stages to capture increasingly abstract features as the network deepens [54].

CNNs have been widely adopted in computational pathology, excelling in tasks such as tissue classification from patches, nuclei detection, and segmentation of histological structures in H&E-stained images, where local patterns and texture variations are crucial for diagnosis (as discussed in Section II-A1). However, while highly effective at capturing local information, traditional CNN architectures inherently possess limited receptive fields. This can make it challenging for them to model long-range spatial dependencies and global contextual information across extensive regions of WSIs without resorting to specialized multi-stage processing or patch aggregation strategies. These limitations in capturing broader context, especially relevant for understanding complex tissue architectures (a key challenge mentioned in Section II-A2), have partly motivated the exploration of alternative architectures, such as ViTs, and the development of hybrid models.

### C. VISION TRANSFORMERS

ViTs are a class of DL models adapted from the Transformer architecture originally designed for natural language processing. Unlike CNNs, which rely on local receptive fields and shared weights to extract hierarchical spatial features, ViTs model image data as sequences and learn global dependencies through self-attention mechanisms. This approach allows them to capture long-range contextual relationships from the beginning of the network, without a locality-based inductive bias [17].

The standard ViT architecture divides an image into non-overlapping patches and treats each patch as a token, similar to words in a sentence. Each token is linearly embedded and enriched with positional information before being fed into a stack of Transformer encoder blocks. These blocks consist of multi-head self-attention (MSA) and feed-forward layers (MLP), each surrounded by normalization and residual connections. The final output token representations are

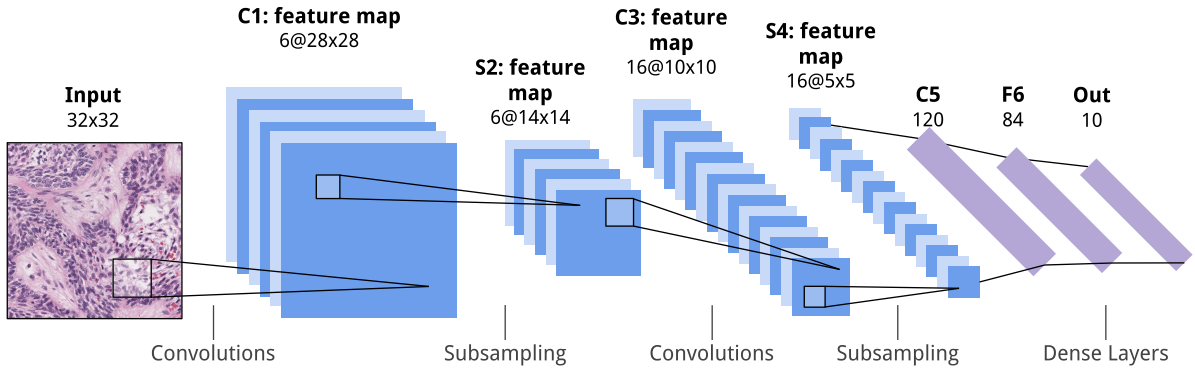



**FIGURE 3.** A standard CNN architecture. Input images are processed through a series of convolutional and pooling layers to extract hierarchical features. Adapted from LeCun et al. [53].

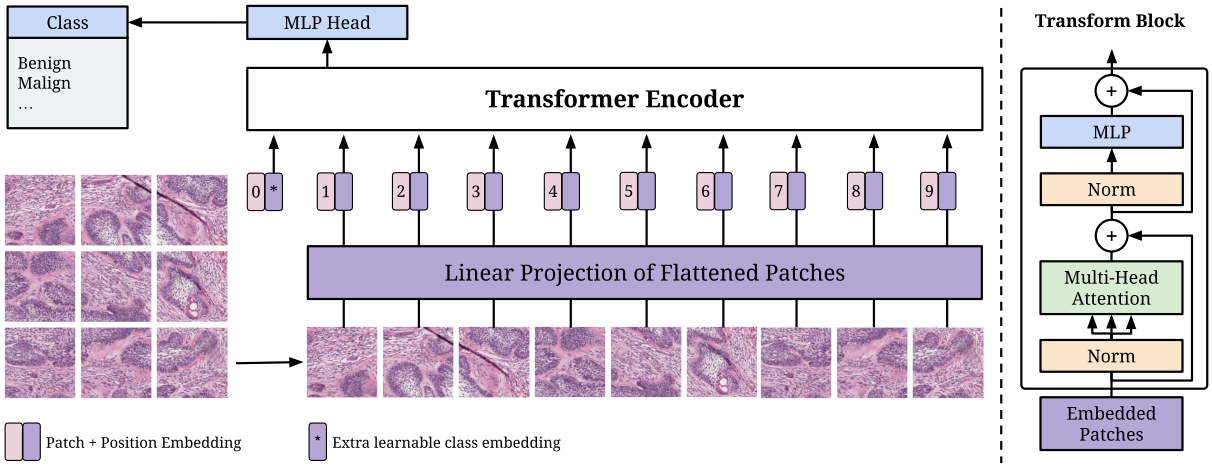



**FIGURE 4.** A ViT architecture for image classification. The input image is divided into fixed-size patches, which are flattened and linearly projected. A learnable class embedding is prepended to the sequence, and position embeddings are added. The sequence is passed through multiple Transformer encoder blocks. The final class prediction is made via an MLP head using the class token output. Adapted from Dosovitskiy et al. [17].

aggregated, often using the representation of a dedicated [CLS] token, and passed to a classifier (see Fig. 4).

Let $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$ be an RGB input image. It is divided into $N$ non-overlapping patches of size $P \times P$, resulting in $N = \frac{HW}{P^2}$ patches. Each patch is flattened and projected into a $D$-dimensional embedding space using a trainable linear projection $\mathbf{E} \in \mathbb{R}^{(P^2\cdot 3)\times D}$. Let $\mathbf{X}_0 \in \mathbb{R}^{(N+1)\times D}$ be the input sequence to the Transformer, where the first token is a learnable class token $\mathbf{x}_{\text{cls}}$ and the remaining tokens are the embedded image patches. Positional embeddings $\mathbf{P} \in \mathbb{R}^{(N+1)\times D}$ are added to preserve spatial information:

$$\mathbf{X}_0 = [\mathbf{x}_{\text{cls}}; \mathbf{x}_1; \ldots; \mathbf{x}_N] + \mathbf{P}.$$

Each Transformer encoder layer updates $\mathbf{X}_l$ by applying MSA followed by a feed-forward MLP:

$$\mathbf{X}'_l = \text{MSA}(\text{LN}(\mathbf{X}_{l-1})) + \mathbf{X}_{l-1},$$
$$\mathbf{X}_l = \text{MLP}(\text{LN}(\mathbf{X}'_l)) + \mathbf{X}'_l,$$

where LN denotes layer normalization and residual connections are employed after each block. The final prediction is obtained by applying a classification head to the output of the [CLS] token.

Since the introduction of the original ViT in [17], several improvements have been proposed. DeiT (Data-efficient Image Transformer) introduced knowledge distillation techniques for training ViTs with fewer data [55]. Swin Transformer proposed a hierarchical structure with shifted windows, enabling local attention while maintaining scalability to high-resolution inputs [56]. Other notable variants include T2T-ViT (Tokens-to-Token Vision Transformer) [57], PVT (Pyramid Vision Transformer) [58], CvT (Convolutional vision Transformer) [21], and MobileViT [59], each addressing limitations in data efficiency, computational complexity, or spatial resolution handling.

In the domain of histopathology, ViTs have shown promise due to their ability to model global tissue structures and spatial relationships. This is particularly useful in WSIs, where local features alone may be insufficient to

distinguish between subtle pathological patterns. ViTs allow modeling of contextual dependencies across regions of a slide, which is essential for robust and explainable diagnosis in computational pathology [19], [27]. In particular, ViTs are well-suited for identifying architectural distortions and tissue organization patterns—key indicators in H&E-stained slides for grading and prognosis. Their ability to integrate global contextual information from the start enables more holistic tissue assessment compared to traditional CNNs. Nonetheless, the absence of locality priors and the high data requirements of ViTs can limit their standalone effectiveness in histopathology. This has led to the emergence of hybrid CNN-Transformer architectures that aim to combine the strengths of both paradigms, as explored in the following sections.

### D. HYBRID CNN-TRANSFORMERS

Hybrid CNN-Transformer architectures exhibit diverse module organization and integration patterns, making it challenging to compare studies directly. The literature presents two main taxonomies for these hybrid architectures: one proposed by Khan et al. [28] and another by Kim et al. [24]. Both aim to categorize the different approaches used to combine CNNs and ViTs.

Focusing on integration strategies, Khan's taxonomy [28] provides a more granular classification of CNN-ViT hybrids by introducing multiple levels of integration, including early-layer, lateral-layer, sequential, parallel, block, hierarchical, attention-based, and channel boosting integration. This taxonomy offers a detailed perspective on CNN-Transformer hybridization by highlighting different structural adaptations and their respective advantages. Fig. 5a provides details of Khan's taxonomy.

Kim's taxonomy [24] classifies hybrid architectures based on their structural design (see Fig. 5b) and integration strategies (see Fig. 5c). It identifies two main architectural designs: parallel and sequential. In the parallel design, CNN and ViT operate independently, with their outputs fused through mechanisms like cross-attention. In contrast, in the sequential design, the data is processed through CNNs and then the extracted features are used in the ViTs, or in the reverse flow where ViT-derived representations are subsequently processed by CNN modules, creating a higher interdependence between the two modules. Additionally, Kim's taxonomy categorizes merging strategies into feature reshaping, positional encoding, and fusing modules, while distinguishing models based on their transformer utilization—either as an encoder for feature embedding or a decoder for generating outputs.

While the taxonomies proposed by Khan et al. [28] and Kim et al. [24] offer valuable structural insights, Kim focuses on architectural layouts and fusion strategies, whereas Khan emphasizes integration at specific network layers. This review introduces a complementary taxonomy grounded in the functional requirements of key computer vision tasks (see Fig. 6).

Rather than solely categorizing CNN–ViT associations, our approach emphasizes how hybrid architectures are designed and optimized for classification, segmentation, and object detection. By aligning architectural choices with task-specific demands, this task-oriented taxonomy offers a more practical and clinically relevant perspective on the use of hybrid models in histopathological image analysis.

For classification tasks, architectures typically consist of three main components: an encoder for feature extraction (using convolutional layers in CNNs or self-attention mechanisms in Transformers), a feature adaptation stage that refines extracted features before classification, and a final classifier (see Fig. 6a). Segmentation tasks, on the other hand, are structured around four primary components: the encoder, the bottleneck, the decoder, and skip connections that help recover spatial information lost during downsampling (see Fig. 6b). Object detection architectures follow another distinct design, incorporating a backbone for feature extraction, a neck for multi-scale feature fusion and a head for generating predictions (see Fig. 6c).

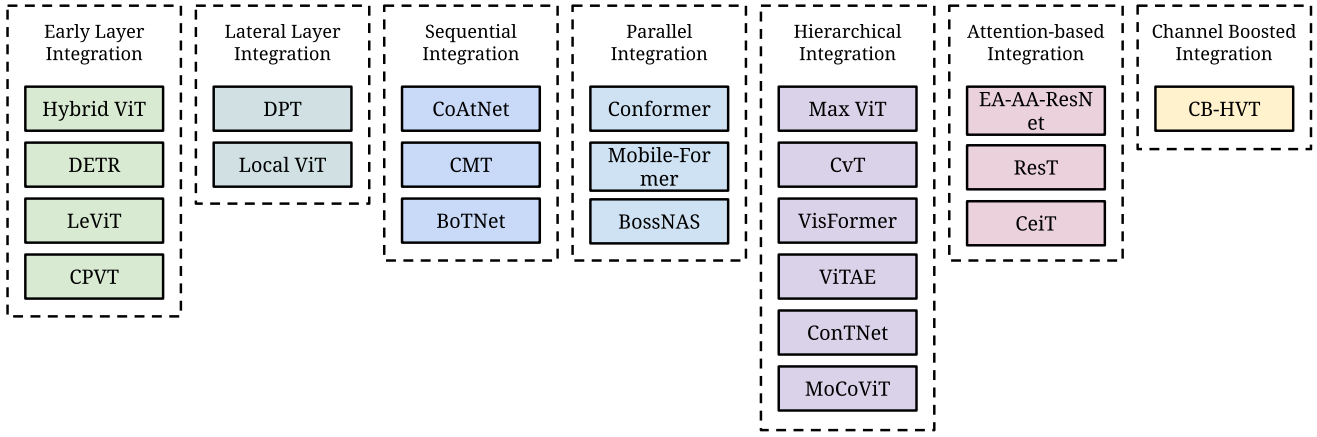
While this review is scoped to histopathology, the proposed task-oriented taxonomy is inherently generalizable. The core tasks of classification, segmentation, and detection are fundamental challenges in computer vision, ones that transcend specific domains. We identify standard architectural blueprints: encoder-classifier structures are used for classification, encoder-decoder architectures with skip-connections for segmentation, and backbone-neck-head designs for object detection. This framework is directly applicable to other medical imaging modalities, such as segmenting tumors in magnetic resonance imaging (MRI) or detecting nodules in computed tomography (CT) images [60], [61], [62]. Furthermore, its applicability extends beyond the medical field to general-purpose computer vision, such as in semantic segmentation for satellite imagery [63], [64]. In this and other general-purpose domains, hybrid models are built upon these same task-specific principles.

The following sections describe each of these categories in detail, highlighting key design choices and representative architectures.

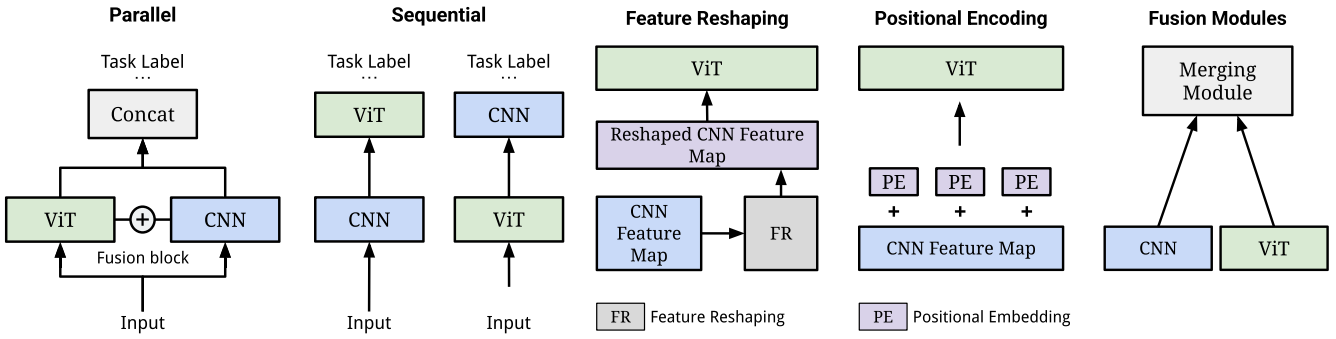
#### 1) HYBRID ARCHITECTURES FOR IMAGE CLASSIFICATION

Classification models aim to assign a single or multiple labels to an image, making them the simplest form of image analysis. The CNN-based and Transformer-based architectures, from early models like LeNet-1 [65] and AlexNet [66] to more recent ones like ResNet [67], VGG [68], ViT [17], and Swin Transformer [69], generally follow a common structure in classification tasks. They consist of three components: an encoder for feature extraction, a feature adaptation stage and a classifier layer. Table 2 presents a comparative analysis of these components in classic architectures.

Based on this, we present the categorized hybrid models for classification using three key facets:




(a) Khan's taxonomy – Integration strategies.




(b) Kim's taxonomy – Architecture variations.

(c) Kim's taxonomy – Merging strategies variations.

**FIGURE 5.** The hybrid CNN–Transformer taxonomies adapted from Khan et al. [28] and Kim et al. [24].

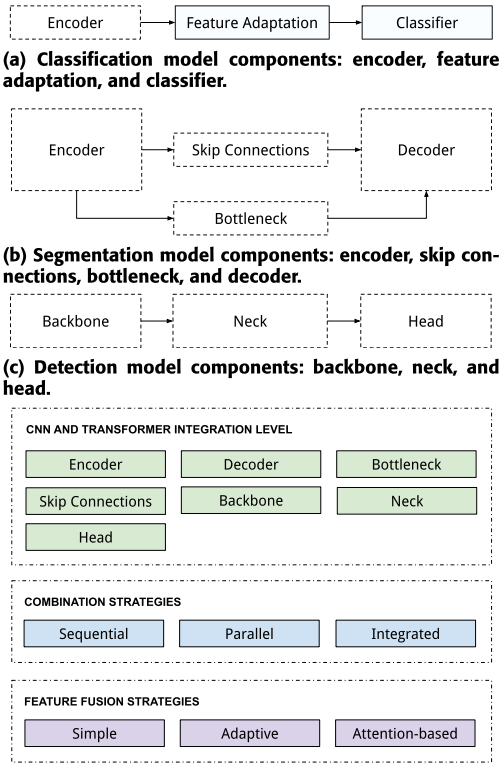
**TABLE 2.** Comparison of CNN-based and Transformer-based architectures for image classification.

| Architecture | Encoder | Feature Adaptation | Classifier |
|---|---|---|---|
| LeNet-1 [65] | Conv layers | Flatten | Fully connected |
| LeNet-5 [53] | Conv layers + Tanh | Flatten | Fully connected |
| AlexNet [66] | Conv layers + ReLU | Flatten | Fully connected |
| VGG [68] | Conv layers + ReLU | Flatten | Fully connected |
| ResNet [67] | Conv layer + Residual Blocks | Global Average Pooling | Fully connected |
| MobileNet [70] | Depthwise Separable Conv + ReLU | Global Average Pooling | Fully connected |
| EfficientNet [71] | MBConv + Swish Activation | Global Average Pooling | Fully connected |
| ViT [17] | Patch Embedding + Transformer Encoder | Token CLS | Fully connected |
| Swin Transformer [69] | Patch Embedding + Swin Blocks | Global Average Pooling | Fully connected |
| CoAtNet [72] | CoAtNet blocks | Global Average Pooling | Fully connected |
| Med-Former [73] | Patch Embedding + Med-Former blocks | Token CLS | Fully connected |

- CNN and Transformer Integration Level: Specifies at which stage CNN and Transformer are incorporated within the network.
- Combination Strategy: Defines the structural approach applied to integrate CNNs and Transformers within a hybrid model.
- Feature Fusion Strategies: Determines how information from CNN and Transformer components is combined.

The integration of CNN and Transformer components in hybrid architectures depends on their placement within the network. These components are commonly used at the encoder level, where they contribute to initial feature extraction (see Fig. 6). However, in the feature adaptation and classifier level, CNNs or Transformers are generally not employed, as these steps primarily involve operations such as feature refinement through pooling or fully connected layers. As shown in Table 2, the majority of architectures rely on feature adaptation techniques like global average pooling or token classification at these levels, rather than additional CNN or Transformer components.

The combination strategy refers to how CNNs and Transformers are structurally integrated within a hybrid model. This strategy influences how features are extracted,




**(a) Classification model components: encoder, feature adaptation, and classifier.**

**(b) Segmentation model components: encoder, skip connections, bottleneck, and decoder.**

**(c) Detection model components: backbone, neck, and head.**

**(d) Proposed taxonomy: CNN/Transformer integration levels, combination strategies, and feature fusion strategies.**

**FIGURE 6. Diagram illustrates different components of deep learning models and a proposed taxonomy.**

combined, and refined. In general, we identify three common approaches:

- Sequential: CNNs are used first to extract local features from the input image. These feature maps are then passed to a Transformer module, which refines the representations by capturing long-range dependencies and global context. This approach benefits from the inductive biases of CNNs while enhancing feature richness through self-attention.
- Parallel: CNNs and Transformers operate on the same input data simultaneously but in separate branches. Each branch processes the input independently, and their outputs are later fused, either by concatenation, addition, or another fusion technique. This setup allows the model to learn complementary representations at different abstraction levels.
- Integrated: Convolution and self-attention mechanisms are combined within the same module or layer. In this case, both local and global features are learned jointly and interactively throughout the network. This design aims to achieve a fine-grained balance between efficiency and expressive power.

These strategies are illustrated in Fig. 6d, and each brings different trade-offs in terms of complexity, performance, and adaptability to specific tasks.

Feature fusion strategies determine how information is combined between the CNN and Transformer components. In this category, there are simple, adaptive, and attention strategies. In the simple fusion methods, such as concatenation and summation, they fixedly combine extracted features without dynamic weighting. Adaptive fusion approaches, including gated fusion and weighted summation, learn to adjust the contribution of each component based on the input data. Attention-based fusion strategies, such as self-attention and cross-attention, leverage attention mechanisms to selectively enhance relevant features before integration (see Fig. 6d).

### 2) HYBRID ARCHITECTURES FOR IMAGE SEGMENTATION

Segmentation models assign a label to each pixel in an image, requiring a more complex architecture than classification models. Hybrid models for segmentation, particularly those inspired by the U-Net architecture [74], can be systematically classified based on their structural components. The U-Net architecture consists of four primary parts: the encoder, the bottleneck, the decoder, and the skip connections (see Fig. 6b). Each part can integrate CNNs, Transformers, or a combination of both, leading to diverse hybrid architectures. Table 3 presents a comparative analysis of these components in well-known segmentation models.

To categorize hybrid models designed for segmentation, in this work, we adopt the same taxonomy used for classification, adjusting it to the segmentation-specific network structure (see Fig. 6d). In this case, the CNN and Transformer integration levels, the combination strategy, and the feature fusion strategies apply to the four major components of a U-Net-based architecture: the encoder, the bottleneck, the decoder, and the skip connections (Fig. 6b and Fig. 6d).

### 3) HYBRID ARCHITECTURES FOR IMAGE DETECTION

Object detection models aim to identify and localize multiple objects within an image. This task requires a more complex pipeline than classification or segmentation. These models typically consist of three primary components: a backbone for feature extraction, a neck for multi-scale feature fusion, and a head for bounding box and class prediction (Fig. 6c). In hybrid architectures, CNNs or Transformers can be integrated into any of these components, leading to diverse model designs.

Traditional object detection models, such as Faster R-CNN [81] and RetinaNet [82], rely on CNN-based backbones (e.g., ResNet-50) with feature fusion via FPN and anchor-based heads, employing Non-Maximum Suppression (NMS) for post-processing. The YOLOv5 model [83] optimizes efficiency with CSPDarknet and an anchor-free (or anchor-based, depending on the version) head, while EfficientDet refines feature reuse with BiFPN. In contrast, DETR [84] integrates a Transformer encoder with a CNN backbone, eliminating the need for FPN and NMS through set-based prediction with Hungarian matching. Table 4 summarizes these architectures and their key components.

**TABLE 3.** Comparison of U-Net-based architectures.

| Architecture | Encoder | Bottleneck | Decoder | Skip Connections |
|---|---|---|---|---|
| U-Net [74] | Conv layers | Conv layers | Conv layers | Direct |
| UNet++ [75] | Conv layers | Conv layers | Conv layers | Direct (with nested connections) |
| Attention U-Net [76] | Conv layers | Conv layers | Conv layers | Refined (with attention gates) |
| TransUNet [77] | Conv layers + ViT | Transformer | Conv layers | Direct |
| Swin-Unet [78] | Transformer | Transformer | Transformer | Transformer-enhanced |
| MedT [79] | Conv layers | Transformer | Conv layers + ViT | Transformer-enhanced |
| UCTransNet [80] | Conv layers | Conv layers | Conv layers | Transformer-enhanced |

**TABLE 4.** Analysis of object detection hybrid architectures.

| Architecture | Backbone | Neck | Head | Post-Processing |
|---|---|---|---|---|
| Faster R-CNN [81] | ResNet-50/101 | FPN | RPN + FC | NMS |
| YOLOv5 [83] | CSPDarknet | CSP-PAN | Anchor-free/based | NMS |
| RetinaNet [82] | ResNet-50/101 | FPN | Anchor-based | NMS |
| DETR [84] | ResNet-50/101 | None | Transformer | Hungarian Matching |
| EfficientDet [85] | EfficientNet | BiFPN | Anchor-based | NMS |

**TABLE 5.** Inclusion criteria for the scoping review.

| Criteria | Description |
|---|---|
| Study Design | Peer-reviewed journal articles on hybrid CNN-Transformer models for histopathology. |
| Publication Period | Studies published from 2020 onward, marking the emergence of ViTs. |
| Language | Studies written in English for consistency in data extraction and interpretation. |
| Domain and Context | Focus on histopathology using hybrid CNN-Transformer models for segmentation, detection, or classification. |
| Model Type | Only models combining CNN and Transformer components were included; standalone models were excluded. |
| Availability of Results | Must detail model architecture, training process, evaluation metrics, and datasets used. |

Hybrid object detection models follow the same taxonomy as classification and segmentation, adapted to their structure (see Fig. 6d). The CNN and Transformer models' integration levels, combination strategies, and feature fusion methods apply to the backbone, neck, and head, which handle feature extraction, fusion, and object prediction (see Fig. 6c).

## III. METHOD

This scoping review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) guidelines [86]. Additionally, the methodology was guided by the framework established by the Joanna Briggs Institute (JBI) for conducting scoping reviews [32].

### A. PROTOCOL AND REGISTRATION

The protocol for this review was developed a priori and revised by the research team to ensure methodological rigor and transparency. It outlines the eligibility criteria, data sources, search strategy, study selection process, and data extraction methods. The protocol has been registered in the Open Science Framework (OSF) and is accessible at https://doi.org/10.17605/OSF.IO/A8C5D to promote reproducibility and allow for external validation.

### B. ELIGIBILITY CRITERIA

The eligibility criteria for this scoping review were defined using the population, concept and context (PCC) framework [32] to ensure the inclusion of relevant studies while maintaining methodological rigor. The population (P) consists of histopathology images, the concept (C) focuses on hybrid CNN-Transformer models, and the context (C) includes tasks related to histopathology image analysis, such as segmentation, detection, and classification. The inclusion criteria used in this scoping review are summarized in Table 5. The exclusion criteria used in this scoping review are outlined in Table 6.

These criteria were designed to focus the review's scope and ensure the quality and relevance of the extracted data. The decision to include only peer-reviewed journal articles and exclude conference papers aims to ensure that all included studies feature complete methodologies and validated results, rather than preliminary findings. The publication period (2020 onward) was selected to align with the emergence and expansion of ViTs in the literature.

The emphasis on hybrid models aligns directly with the Concept of the PCC framework, enabling a focused analysis of the specific synergy between CNNs and Transformers. Finally, criteria requiring sufficient methodological details and excluding studies that lack them were crucial for enabling the detailed data extraction and synthesis. The PCC framework guided this entire process to ensure all criteria aligned with the review's objectives.

**TABLE 6.** Exclusion criteria for the scoping review.

| Criteria | Description |
|---|---|
| Non-Hybrid Models | Studies focusing solely on traditional CNNs or standalone Transformers without hybridization. |
| Study Type | Conference papers, reviews, editorials, or opinion pieces were excluded. |
| Insufficient Methodology | Studies lacking sufficient details on model architecture, evaluation metrics, or dataset specifications. |

### C. INFORMATION SOURCES

The literature search was conducted in four major databases: Web of Science, IEEE Xplore, Scopus, and Medline (PubMed). These sources were chosen for their relevance to medical imaging, artificial intelligence, and computational pathology. To ensure completeness, the reference lists of included studies were screened for additional relevant articles. The last search was executed on February 7, 2025.

### D. SEARCH STRATEGY

A structured search strategy was designed to maximize the retrieval of relevant studies on hybrid CNN-Transformer models for histopathology image processing. The search terms included combinations of keywords and controlled vocabulary (e.g., MeSH terms) related to deep learning, convolutional neural networks, transformers, and histopathology image analysis. Boolean operators (AND, OR) and truncation techniques were applied to optimize sensitivity and specificity.

An example of the search strategy used in Web of Science is presented in Table 7. The complete search strategies for all databases are detailed in the Subsection III-A.

### E. SELECTION OF SOURCES OF EVIDENCE

The selection of sources of evidence followed a two-stage screening process: (1) title and abstract screening and (2) full-text review. All identified citations were exported and uploaded to Rayyan [87], a web-based tool designed for systematic reviews, to facilitate collaboration and minimize bias. Two independent reviewers screened all titles and abstracts for relevance. Studies deemed potentially eligible proceeded to the full-text review stage, where 64 articles were assessed by both reviewers. The inter-reviewer agreement for this stage, measured by Cohen's kappa, was $\kappa = 0.93$, indicating almost perfect agreement [88]. All disagreements at both stages were resolved through discussion or, when consensus was not reached, by consulting a third reviewer.

### F. DATA CHARTING PROCESS

A structured data charting form was developed to extract relevant information from the included studies. Two independent reviewers piloted the form on a subset of studies to ensure consistency and refine data extraction criteria. The form was iteratively updated based on team discussions. Data were independently charted by two reviewers using Google Sheets. Discrepancies were resolved through discussion or, when necessary, consultation with a third reviewer. Extracted data included study characteristics, model architecture, datasets used, evaluation metrics, and key findings.

**TABLE 7.** Example of search strategy executed in Web of Science.

| | |
|---|---|
| Database | Web of Science |
| Search Conducted On | 07/02/2025 |
| Number of Records Retrieved | 544 |
| Search String | TS=("CNN-transformer*" OR "convolution transformer*" OR "hybrid transformer*" OR "CNN-ViT hybrid*" OR "vision transformer*" OR "ViT" OR "visual transformer*" OR "transformer-based model*" OR "self-attention network*" OR "transformer neural network*")<br>AND TS=("histological image*" OR "histopathology image*" OR "tissue image*" OR "medical image*")<br>AND TS=("image classif*" OR "image segment*" OR "feature extract*" OR "object recognit*" OR "pattern recognit*" OR "image analys*")<br>NOT TS=("vitamin" OR "in vitro" OR "vitro")<br>AND PY=(2020-2025) |
| Notes | Search performed in "All Databases". TS denotes a Topic Search, covering title, abstract, author keywords, and Keywords Plus. PY represents the Publication Year filter. The search focuses exclusively on histological and histopathology image analysis tasks. To reduce noise, terms such as "vitamin" and "in vitro" were excluded using the NOT operator. |

### G. DATA ITEMS

The data extracted from the included studies encompassed both qualitative and quantitative aspects relevant to hybrid CNN-Transformer models for histopathology image analysis. Table 8 summarizes the key variables extracted for this scoping review.

A standardized data extraction form was developed and pilot-tested by three reviewers on a subset of included studies to ensure consistency and completeness. The form underwent iterative refinement based on reviewer feedback to address ambiguities and improve clarity. All extracted data were cross-verified for accuracy by at least two reviewers prior to synthesis. No assumptions or extrapolations were made beyond the details explicitly reported in the original studies. The final version of the data extraction form is provided in Subsection III-A.

### H. SYNTHESIS OF RESULTS

The extracted data were synthesized using a combination of narrative descriptions and tabular representations. Studies

**TABLE 8.** Summary of extracted data items for this scoping review.

| Category | Data Items |
|---|---|
| Study Characteristics | Title, year of publication, publication type, venue name, area, country, and funding source. |
| Application Context | Domain, context of application, designed for histopathology, detection, segmentation, classification, main task, method, and main task output. |
| Model Architecture | Segmentation: Encoder, bottleneck, decoder, skip connections, combination strategy, feature fusion. Classification: Combination strategy, feature fusion. Detection: Backbone, neck, head. |
| Dataset | Name, image task, image modality, description, objective, public access, number of samples, number of classes, and dataset URL. |
| Model Training | Optimizer, learning rate scheduler, weight initialization method, warmup strategy, early stopping, data augmentation techniques, and task-specific augmentation. |
| Model Interpretability and Explainability | Description of interpretability and explainability techniques. |
| Evaluation Strategy | Cross-validation, cross-validation details, holdout validation, holdout validation details, stratified splits, external validation, and metrics. |
| Limitations | Study limitations, model limitations, and ethical considerations. |
| Outcomes | Impact of study, practical applications, follow-up experiments, and future work or recommendations. |

were grouped based on their primary tasks (segmentation, classification, or detection), model architecture, dataset usage, and evaluation metrics. Relevant characteristics, such as transformer integration strategies, training methodologies, and performance metrics, were summarized in comparative tables.

No meta-analysis or statistical pooling of results was performed, as this review aimed to provide an overview of the existing literature rather than a quantitative synthesis.

## IV. RESULTS

The scoping review process, outlined in Fig. 7, included 39 studies after thorough screening and eligibility assessment. These studies focused primarily on segmentation, classification, and detection tasks. A comprehensive list of these 39 included articles, categorized by their primary task, is presented in Tables 9–11.

### A. OVERVIEW OF SELECTED STUDIES

The segmentation studies, summarized in Table 9, predominantly utilized fully supervised methods, highlighting the availability of labeled datasets in this domain. Parallel and sequential combination strategies were the most common approaches, emphasizing the need for structured feature flow in segmentation tasks. Notably, self-supervised methods were also investigated, demonstrating the potential for reducing reliance on annotated data [25], [89], [90]. Feature fusion strategies varied, with attention-based fusion frequently used in parallel architectures, whereas sequential models often omitted explicit fusion mechanisms. These findings indicate a preference for direct feature propagation or enhanced multi-scale integration, depending on the segmentation context. In addition, the predominance of fully supervised approaches (73.33%) reflects both the maturity and limitations of current datasets. Semi-supervised and self-supervised models, though fewer, signal a promising direction for addressing annotation scarcity. Their adoption, however, remains limited, largely due to the lack of standardized benchmarks and the difficulty in validating model performance without extensive expert annotations. Future efforts should prioritize the development of robust validation protocols and curated datasets to support the broader implementation of weakly supervised learning strategies in histopathological image analysis.

As shown in Table 10, most classification studies rely heavily on fully supervised methods, with sequential combination strategies being the most commonly used. Weakly supervised approaches were also present, primarily leveraging attention-based fusion to compensate for limited labeled data [37], [103].

Table 11 summarizes two detection studies. Both adopt fully supervised training. The combination strategies differ, with one employing a parallel hybrid design and the other an integrated design. In both cases, feature fusion is attention-based.

Table 12 presents the characteristics and metadata of the included studies, encompassing publication year, country, funding status, and field of publication. The included studies span the years from 2022 to 2025. Most of the studies were published in 2024 (58.97%, n=23), followed by 2023 with 25.64% (n=10). Smaller proportions were published in 2025 (10.26%, n=4) and 2022 (5.13%, n=2).

Fig. 8(a-d) provides an overview of these distributions, illustrating the trends in publication year, journal areas, geographic contribution, and funding status.

Regarding publication type, a substantial proportion of the included studies were published in multidisciplinary journals (56.41%, n=22), reflecting the cross-domain relevance of hybrid CNN–Transformer models in histopathological image analysis. This was followed by publications in computing-focused journals (23.08%, n=9), medical and biomedical journals (17.95%, n=7), and a small number in instrumentation-oriented venues (2.56%, n=1), indicating a predominance of technical dissemination over domain-specific clinical outlets.

In terms of geographic distribution, the majority of studies originated from China (53.85%, n=21), underscoring its leading role in DL research applied to digital pathology. Other notable contributors included India and Iran (each

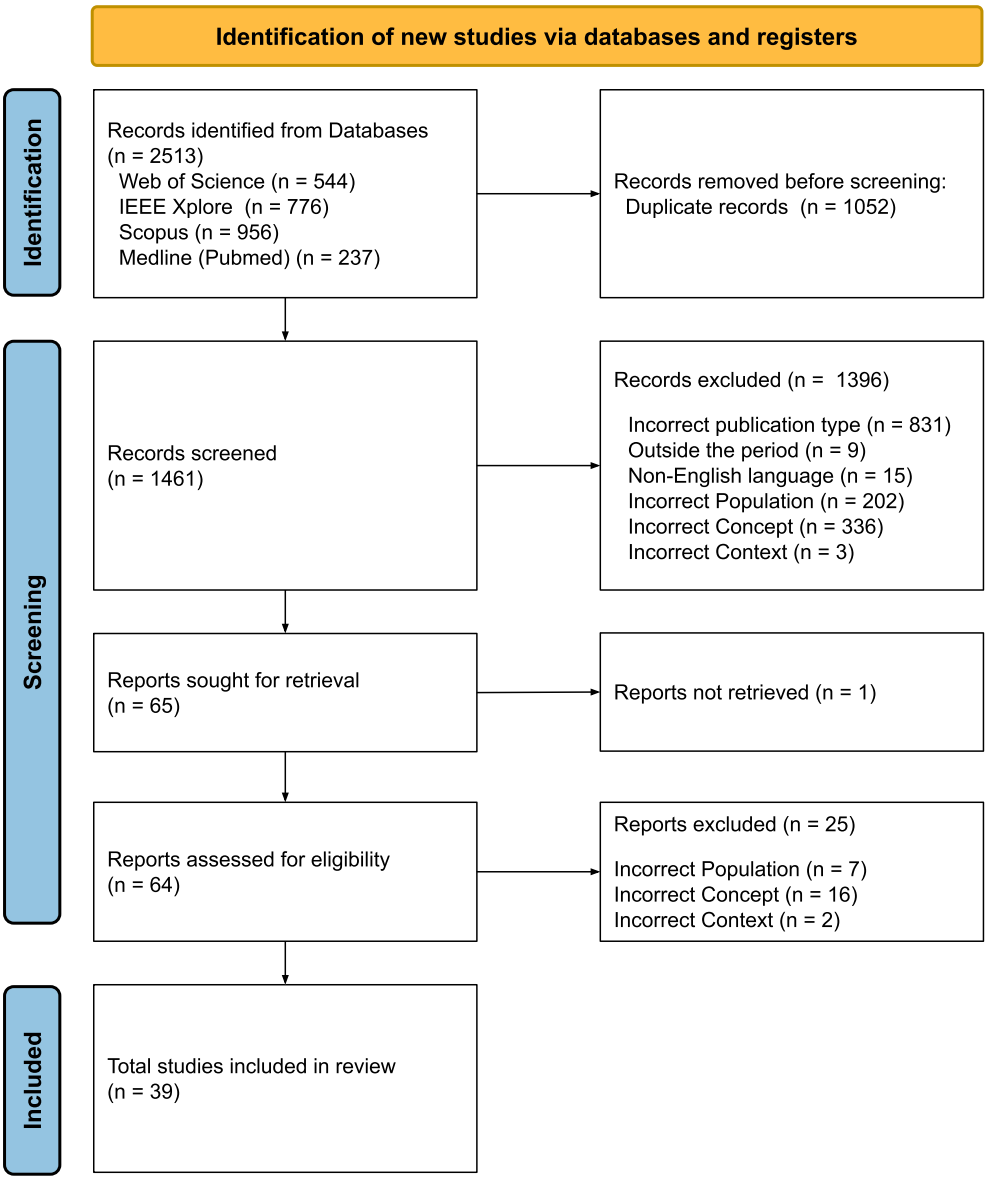



**FIGURE 7.** PRISMA flow diagram illustrating the scoping review process.

with 7.69%, n=3), followed by Pakistan, Turkey, the United States, and the United Kingdom (each contributing 5.13%, n=2). An additional group of studies (10.24%, n=4) was produced through international collaborations or by authors affiliated with institutions in other countries. This geographic pattern reflects both regional research priorities and infrastructure availability for AI-driven medical imaging.

With respect to funding, 51.28% (n=20) of the studies reported receiving financial support, suggesting institutional or governmental investment in this area. In contrast, 5.13% (n=2) explicitly declared the absence of funding, while a significant portion (43.59%, n=17) did not disclose any information regarding funding sources. The lack of transparency in nearly half of the studies may hinder reproducibility and reflect a broader issue in research reporting standards.

### B. CONTEXT OF SELECTED STUDIES

Table 13 provides an overview of the characteristics of the studies included in this review, focusing on their main task, the method used, and whether they were specifically designed for histopathology. The data reveals that classification was the most common task, accounting for 56.41% (n=22) of the studies, followed by segmentation with 38.46% (n=15) and detection (5.13%, n=2).

The predominance of fully supervised approaches (82.05%, n=32) underscores the prevailing reliance on annotated datasets in hybrid CNN-Transformer research

**TABLE 9.** Summary of Studies: Key Tasks, Methods, and Strategies for segmentation.

| Year | Reference | Main Task | Method | Combination Strategy | Feature Fusion Strategy |
|---|---|---|---|---|---|
| 2022 | [89] | Segmentation | Fully Supervised | Parallel | Simple |
| 2023 | [91] | Segmentation | Fully Supervised | Sequential | No feature fusion |
| 2023 | [92] | Segmentation | Semi-Supervised | Parallel | Attention-based |
| 2023 | [25] | Segmentation | Fully Supervised | Parallel | Attention-based |
| 2024 | [93] | Segmentation | Fully Supervised | Sequential | No feature fusion |
| 2024 | [94] | Segmentation | Fully Supervised | Integrated | Simple |
| 2024 | [95] | Segmentation | Fully Supervised | Sequential | No feature fusion |
| 2024 | [96] | Segmentation | Semi-Supervised | Integrated | Simple |
| 2024 | [97] | Segmentation | Fully Supervised | Integrated | Adaptive |
| 2024 | [98] | Segmentation | Fully Supervised | Sequential | No feature fusion |
| 2024 | [99] | Segmentation | Self-Supervised | Parallel | Attention-based |
| 2024 | [100] | Segmentation | Fully Supervised | Sequential | No feature fusion |
| 2024 | [101] | Segmentation | Fully Supervised | Sequential | No feature fusion |
| 2025 | [102] | Segmentation | Fully Supervised | Sequential | No feature fusion |
| 2025 | [90] | Segmentation | Self-Supervised | Parallel | Simple |

**TABLE 10.** Summary of Studies: Key Tasks, Methods, and Strategies for classification.

| Year | Reference | Main Task | Method | Combination Strategy | Feature Fusion Strategy |
|---|---|---|---|---|---|
| 2022 | [104] | Classification | Fully Supervised | Sequential | No feature fusion |
| 2023 | [105] | Classification | Fully Supervised | Parallel | Adaptive |
| 2023 | [36] | Classification | Weakly Supervised | Sequential | No feature fusion |
| 2023 | [106] | Classification | Fully Supervised | Integrated | Attention-based |
| 2023 | [107] | Classification | Fully Supervised | Parallel | Attention-based |
| 2023 | [37] | Classification | Weakly Supervised | Parallel | Attention-based |
| 2023 | [108] | Classification | Fully Supervised | Sequential | No feature fusion |
| 2024 | [23] | Classification | Fully Supervised | Sequential | No feature fusion |
| 2024 | [109] | Classification | Fully Supervised | Sequential | No feature fusion |
| 2024 | [110] | Classification | Fully Supervised | Sequential | No feature fusion |
| 2024 | [111] | Classification | Fully Supervised | Integrated | Simple |
| 2024 | [112] | Classification | Fully Supervised | Sequential | No feature fusion |
| 2024 | [113] | Classification | Fully Supervised | Integrated | No feature fusion |
| 2024 | [114] | Classification | Fully Supervised | Parallel | Simple |
| 2024 | [115] | Classification | Fully Supervised | Parallel | Simple |
| 2024 | [116] | Classification | Fully Supervised | Integrated | Adaptive |
| 2024 | [117] | Classification | Fully Supervised | Integrated | Attention-based |
| 2024 | [118] | Classification | Fully Supervised | Sequential | No feature fusion |
| 2024 | [119] | Classification | Fully Supervised | Sequential | No feature fusion |
| 2024 | [120] | Classification | Fully Supervised | Sequential | No feature fusion |
| 2025 | [103] | Classification | Weakly Supervised | Parallel | Attention-based |
| 2025 | [121] | Classification | Fully Supervised | Sequential | No feature fusion |

**TABLE 11.** Summary of Studies: Key Tasks, Methods, and Strategies for detection.

| Year | Reference | Main Task | Method | Combination Strategy | Feature Fusion Strategy |
|---|---|---|---|---|---|
| 2023 | [122] | Detection | Fully Supervised | Parallel | Attention-based |
| 2024 | [123] | Detection | Fully Supervised | Integrated | Attention-based |

for histopathology. Nevertheless, a smaller but growing body of work explores alternative training paradigms, including weakly supervised (7.69%, n=3), semi-supervised (5.13%, n=2), and self-supervised learning (5.13%, n=2). These methods signal a promising direction for mitigating annotation bottlenecks, especially in domains where expert labeling is costly and time-consuming.

Regarding their applicability to histopathology, a significant portion of studies (64.1%, n=25) were explicitly designed for this domain, highlighting the growing interest in leveraging advanced computational methods for histopathology analysis. Conversely, 35.9% (n=14) were not developed exclusively for histopathology images but included experiments involving these images alongside other

**TABLE 12.** Characteristics of the studies used in this review, including year, publication area, country and funding.

| Feature | Studies, n (%) | References |
|---|---|---|
| **Year** | | |
| 2025 | 4 (10.26%) | [90], [102], [103], [121] |
| 2024 | 23 (58.97%) | [23], [93]–[101], [109]–[120], [123] |
| 2023 | 10 (25.64%) | [25], [36], [37], [91], [92], [105]–[108], [122] |
| 2022 | 2 (5.13%) | [89], [104] |
| **Publication area** | | |
| Multidisciplinary | 22 (56.41%) | [23], [25], [36], [93]–[95], [99], [100], [104], [106], [108]–[116], [118], [120], [122] |
| Computing | 9 (23.08%) | [37], [89]–[92], [97], [117], [119], [121] |
| Medical and Biomedical | 7 (17.95%) | [96], [98], [101]–[103], [107], [123] |
| Instrumentation | 1 (2.56%) | [105] |
| **Country** | | |
| China | 21 (53.85%) | [25], [36], [37], [89], [91], [94], [95], [97]–[103], [105], [107], [108], [113], [117], [119], [123] |
| Other | 4 (10.24%) | [23], [106], [118], [121] |
| India | 3 (7.69%) | [90], [110], [120] |
| Iran | 3 (7.69%) | [92], [109], [111] |
| Pakistan | 2 (5.13%) | [115], [122] |
| Turkey | 2 (5.13%) | [114], [116] |
| USA | 2 (5.13%) | [93], [112] |
| United Kingdom | 2 (5.13%) | [96], [104] |
| **Funding** | | |
| Yes | 20 (51.28%) | [23], [36], [37], [89], [91], [93]–[97], [103]–[109], [119], [122], [123] |
| No | 2 (5.13%) | [100], [116] |
| Not Mentioned | 17 (43.59%) | [25], [90], [92], [98], [99], [101], [102], [110]–[115], [117], [118], [120], [121] |

**TABLE 13.** Characteristics of the studies based on main task, method, and histopathology-specific design.

| Feature | Studies, n (%) | References |
|---|---|---|
| **Main Task** | | |
| Classification | 22 (56.41%) | [23], [36], [37], [103]–[121] |
| Segmentation | 15 (38.46%) | [25], [89]–[102] |
| Detection | 2 (5.13%) | [122], [123] |
| **Method** | | |
| Fully Supervised | 32 (82.05%) | [23], [25], [89], [91], [93]–[95], [97], [98], [100]–[102], [104]–[123] |
| Weakly Supervised | 3 (7.69%) | [36], [37], [103] |
| Self-Supervised | 2 (5.13%) | [90], [99] |
| Semi-Supervised | 2 (5.13%) | [92], [96] |
| **Designed for Histopathology** | | |
| Yes | 25 (64.1%) | [23], [36], [37], [96]–[98], [101], [103]–[112], [114]–[116], [118], [120]–[123] |
| No | 14 (35.9%) | [25], [89]–[95], [99], [100], [102], [113], [117], [119] |

imaging modalities, potentially contributing insights relevant to the field.

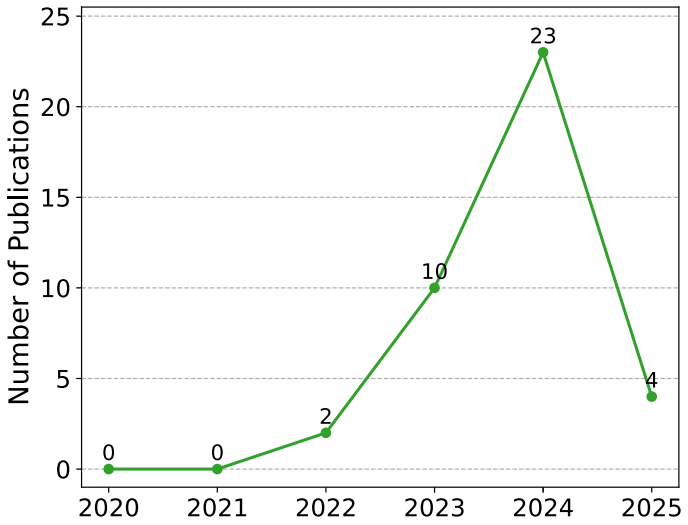
Fig. 9 presents these distributions. Fig. 9a highlights the predominance of classification tasks, while Fig. 9b illustrates the overwhelming use of fully supervised methods. Fig. 9c emphasizes that most studies were designed specifically for histopathology, reinforcing the relevance of this research focus.

### C. HYBRID ARCHITECTURES AND INTEGRATION STRATEGIES

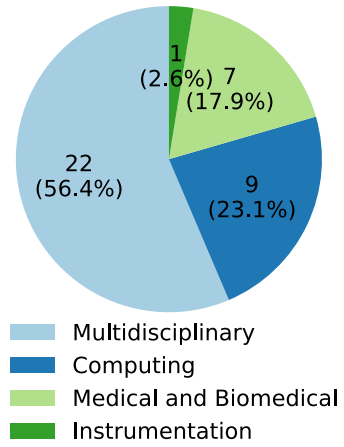
This section categorizes the architectural strategies adopted to integrate CNN and Transformer components in hybrid models for histopathological image analysis. The reviewed studies were grouped into four integration schemes, such as sequential, parallel and integrated, based on how convolutional and attention-based features are combined. We summarize the structural components of each approach and examine their use across classification, segmentation, and detection tasks, with attention to architectural composition and computational trade-offs.

#### 1) SEGMENTATION STUDIES

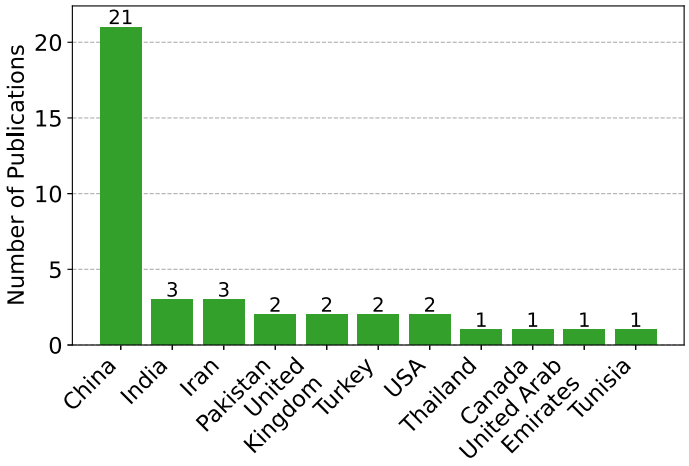
Table 14 provides an overview of the architectural components used in segmentation studies. The distribution of encoder, bottleneck, decoder, and skip connection designs



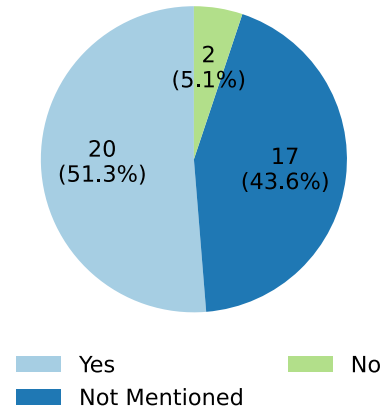

(a) Distribution by year.




(b) Journal areas.




(c) Geographic distribution.




(d) Funding sources.

**FIGURE 8.** Characterization of the included studies.

reflects a growing trend toward hybrid architectures. The encoder structure was predominantly hybrid (53.33%, n=8), followed closely by CNN-based designs (46.67%, n=7). For the bottleneck component, hybrid designs were also the most prevalent (46.67%, n=7), followed by CNN-based (26.67%, n=4) and transformer-based (20.0%, n=3) structures. Among the decoder designs, hybrid structures predominated (53.33%, n = 8), followed by CNN-based structures (40.00%, n = 6). Skip connections were most frequently implemented using custom modules (53.33%, n=8), followed by direct connections (26.67%, n=4) and transformer-based mechanisms (13.33%, n=2), indicating a preference for more sophisticated feature propagation techniques in segmentation models (see Fig. 10). A notable exception to the conventional U-Net architecture is presented by [96]. As detailed in Table 14, this work is distinguished by its omission of the bottleneck, decoder, and skip connections. Instead of an encoder-decoder structure, its approach classifies image patches to generate a final segmentation map. Despite this distinct architecture, the model employs a hybrid encoder for initial feature extraction.

Regarding integration, the combination strategy and feature fusion techniques varied across studies. Sequential combination strategies were the most common (46.67%, n=7), followed by parallel (33.33%, n=5) and integrated approaches (20.00%, n=3). Notably, most models employing a sequential combination did not use explicit feature fusion (46.67%, n=7), as this approach typically propagates feature representations directly from one component to the next without additional processing or merging. In contrast, models using parallel and integrated strategies more frequently incorporated feature fusion techniques, including simple fusion (26.67%, n=4), attention-based fusion (20.00%, n=3), and adaptive fusion (6.67%, n=1). This distribution

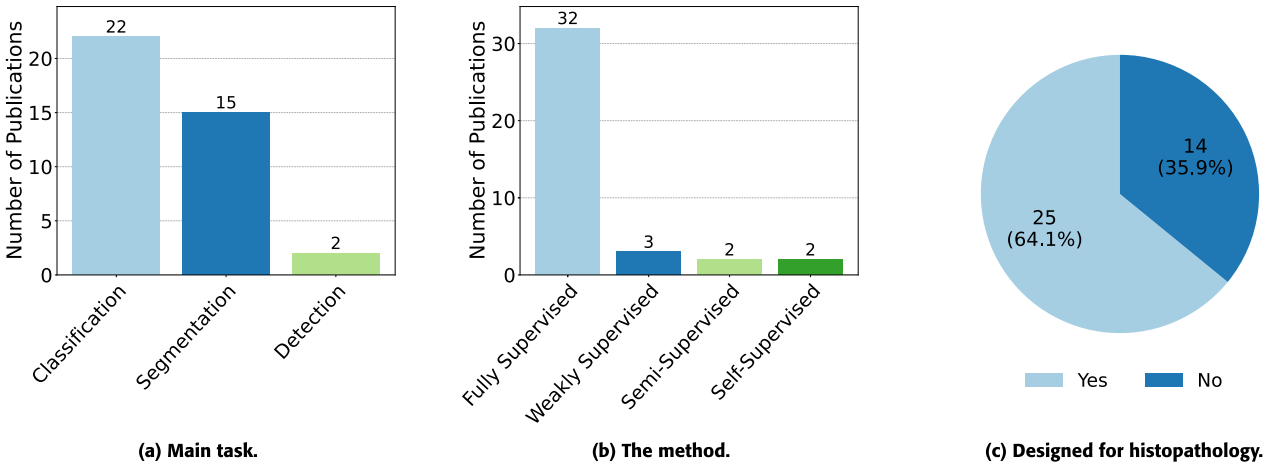



**FIGURE 9.** Distribution of publications based on the application context.

**TABLE 14.** Characteristics of the segmentation studies based on network architecture components.

| Component | Studies, n (%) | References |
|---|---|---|
| **Encoder** | | |
| Hybrid | 8 (53.33%) | [25], [90], [92], [94], [96], [97], [99], [101] |
| CNN | 7 (46.67%) | [89], [91], [93], [95], [98], [100], [102] |
| **Bottleneck** | | |
| Hybrid | 7 (46.67%) | [25], [92], [94], [97], [99], [100], [102] |
| CNN | 4 (26.67%) | [90], [93], [95], [101] |
| Transformer | 3 (20.0%) | [89], [91], [98] |
| None | 1 (6.67%) | [96] |
| **Decoder** | | |
| CNN | 8 (53.33%) | [25], [90], [91], [98]–[102] |
| Hybrid | 6 (40.0%) | [89], [92]–[95], [97] |
| None | 1 (6.67%) | [96] |
| **Skip Connections** | | |
| Custom Module | 8 (53.33%) | [25], [90], [97]–[102] |
| Direct | 4 (26.67%) | [89], [91], [93], [94] |
| Transformer | 2 (13.33%) | [92], [95] |
| None | 1 (6.67%) | [96] |
| **Combination Strategy** | | |
| Sequential | 7 (46.67%) | [91], [93], [95], [98], [100]–[102] |
| Parallel | 5 (33.33%) | [25], [89], [90], [92], [99] |
| Integrated | 3 (20.0%) | [94], [96], [97] |
| **Feature Fusion** | | |
| No feature fusion | 7 (46.67%) | [91], [93], [95], [98], [100]–[102] |
| Simple | 4 (26.67%) | [89], [90], [94], [96] |
| Attention-based | 3 (20.0%) | [25], [92], [99] |
| Adaptive | 1 (6.67%) | [97] |

indicates that explicit feature fusion is more prevalent when multiple feature streams are processed simultaneously, rather than in strictly sequential architectures. These findings underscore the role of feature interaction mechanisms in hybrid architectures, particularly in models designed to enhance representational capacity through multi-stream processing.

One example of a segmentation model is the architecture proposed by Li et al. [91], which employs a CNN-based encoder, applies a transformer in the bottleneck, and utilizes

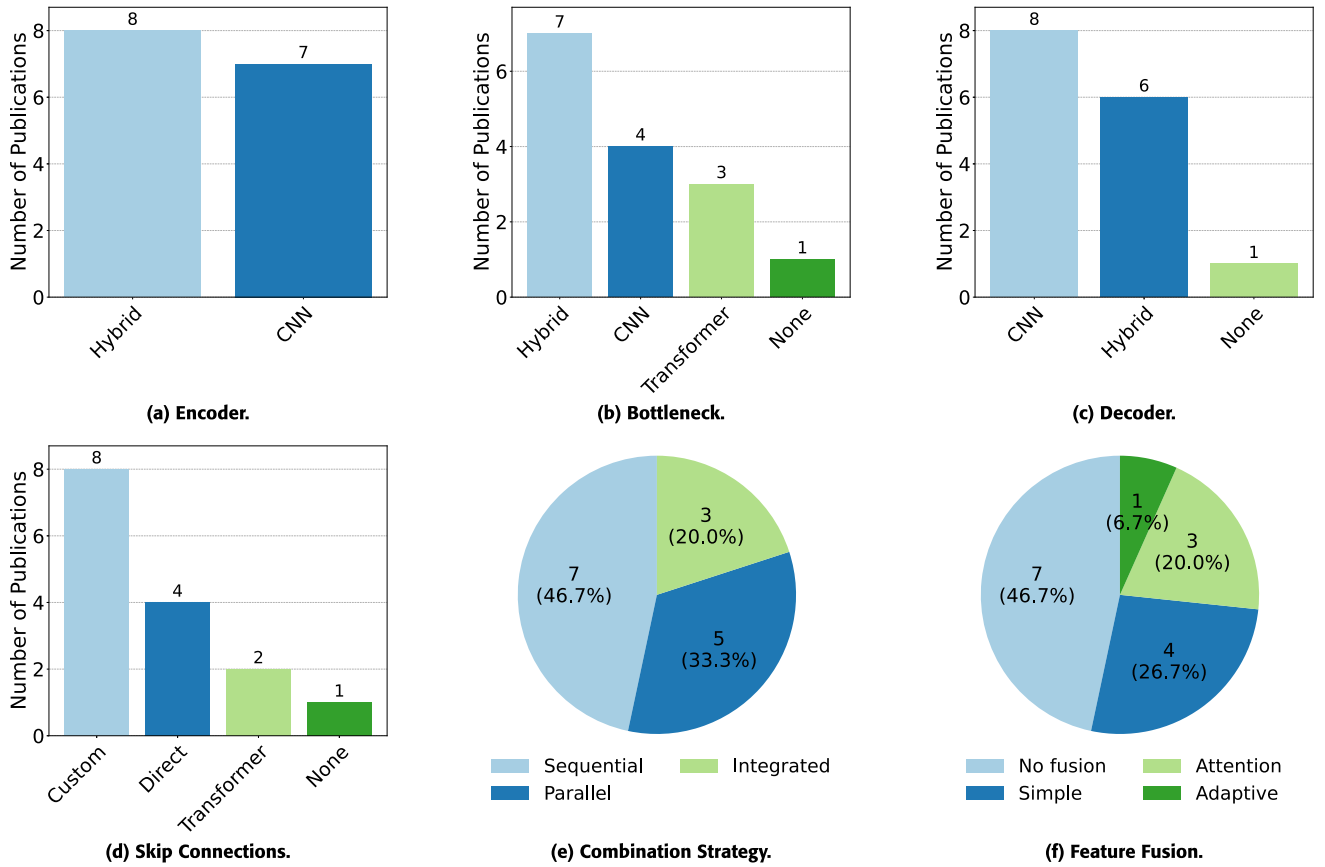



**FIGURE 10.** Distribution of architectural components in segmentation studies. (a) Encoder types and (b) bottleneck structures show a predominance of hybrid designs. (c) Decoder architectures, with CNN-based models being the most frequent. (d) Skip connection strategies highlight a preference for custom implementations. (e) Combination strategies indicate a predominance of sequential approaches. (f) Feature fusion techniques emphasize the use of attention-based and simple fusion, with adaptive fusion remaining less common.

a sequential combination strategy, as illustrated in Fig. 11a. In contrast, the model presented by He et al. [97] adopts a hybrid architecture, integrating CNN and transformer components across all stages, employing an integrated combination strategy, and utilizing adaptive feature fusion, as shown in Fig. 11b.

#### 2) CLASSIFICATION STUDIES

In terms of architectural components, the classification models employed hybrid encoder structures (100.0%, n=22), as expected given the focus of this review (see Fig. 12). While hybrid designs inherently integrate elements from different architectural paradigms, no study applied a distinct CNN-transformer adaptation specifically in the feature adaptation stage.

Regarding integration strategies, sequential combination emerged as the most common approach (50.00%, n=11), followed by parallel (27.27%, n=6) and integrated (22.73%, n=5) strategies. A significant portion of sequentially combined models did not implement explicit feature fusion (54.55%, n=12), as they propagate features through consecutive processing blocks without additional merging. In contrast, attention-based feature fusion was predominantly employed in parallel and integrated architectures (22.70%, n=5), while simple (13.64%, n=3) and adaptive (9.09%, n=2) fusion techniques were less applied. These findings indicate that explicit feature fusion is selectively employed, depending on the degree of architectural modularity and interaction between feature representations (see Table 15).

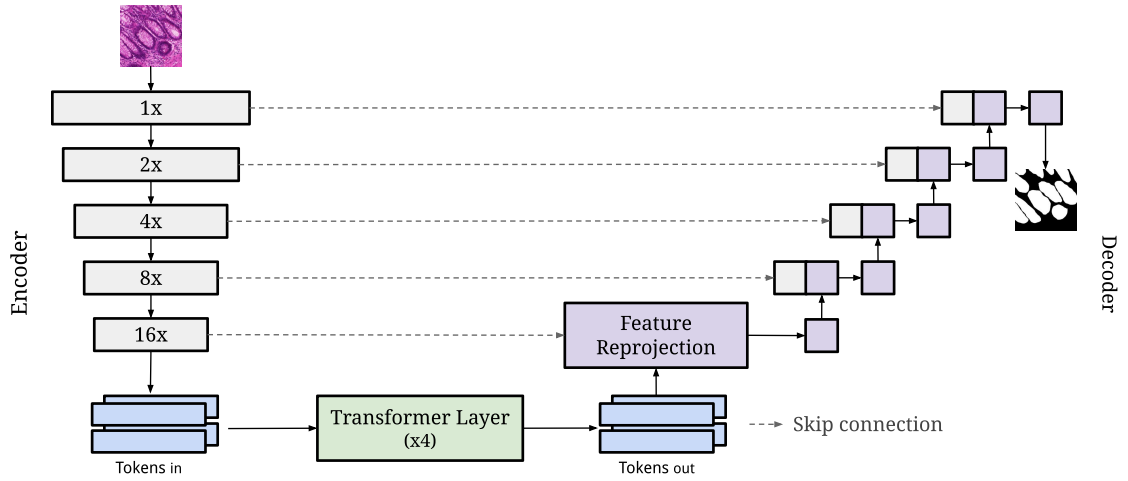
For instance, a classification model is an architecture proposed by Wang et al. [37], which employs a parallel combination strategy, as illustrated in Fig. 13. In contrast, the model presented by Ding et al. [108] utilizes a sequential approach, where a CNN-based encoder is followed by a transformer, as shown in Fig. 13b.

#### 3) DETECTION STUDIES

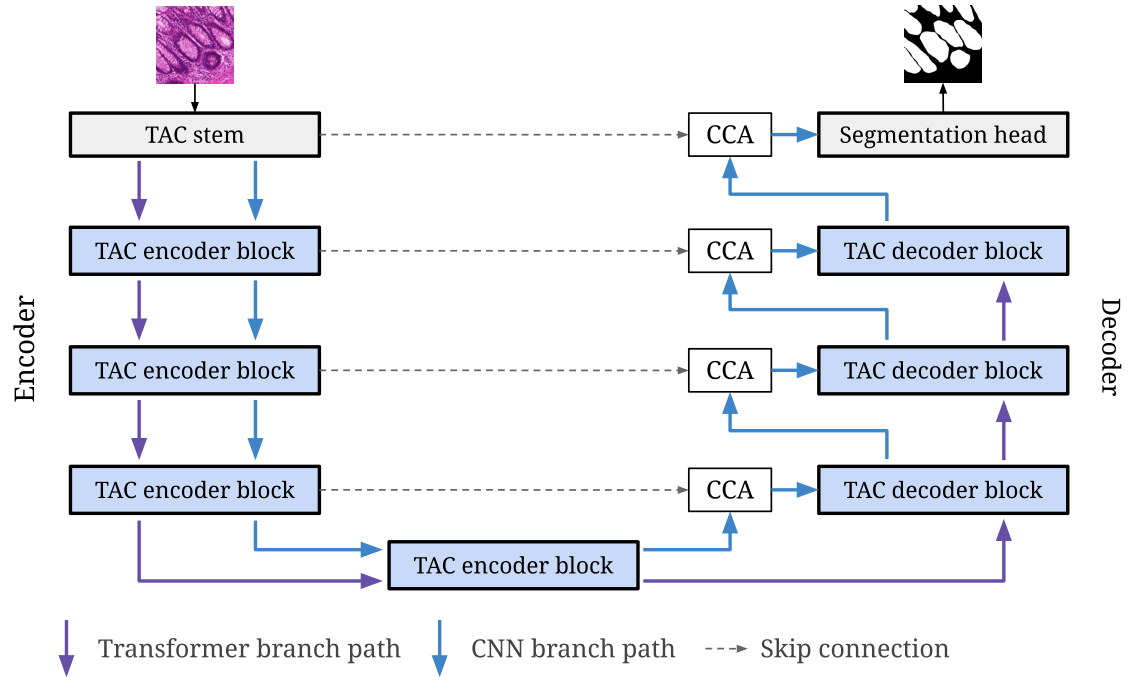
As shown in Table 16, all detection studies employed hybrid backbones (100%, n=2) and CNN-based heads. Neck designs were evenly divided between CNN and hybrid structures. Combination strategies were split between integrated and parallel approaches, and all models incorporated attention-based feature fusion.

### D. DATASETS UTILIZED FOR TRAINING AND EVALUATION

The selection of datasets plays a crucial role in developing and evaluating DL models for medical image analysis. Table 17 summarizes the frequency of dataset usage in




**(a) A CNN-based encoder with a transformer bottleneck and a sequential combination strategy, adapted from Li et al. [92].**




**(b) A hybrid architecture combining CNN and transformer components throughout, utilizing an integrated combination strategy and adaptive feature fusion, adapted from He et al. [98].**

**FIGURE 11. Comparison of segmentation model architectures.**

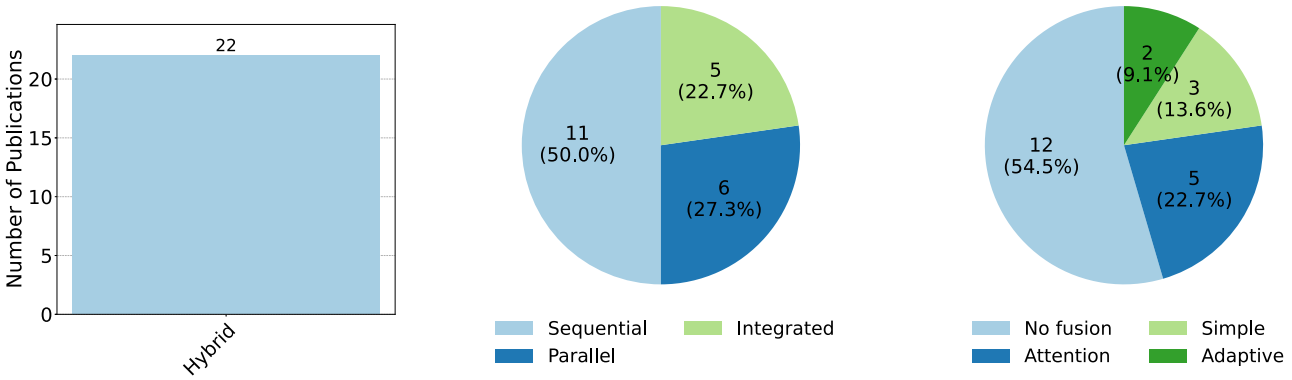
the reviewed studies, categorizing them by task and listing relevant references.

From the analyses presented in this study, it can be seen that the GLAS-2015, BreakHis, MoNuSeg and Bowl-2018 emerge as the most frequently used datasets. These datasets focus primarily on segmentation and classification tasks, highlighting the emphasis on gland, tumor, and nuclei detection in histopathology images. The MoNuSeg dataset, which is specifically designed for nucleus segmentation, is another commonly used resource, reinforcing the importance of precise cell-level annotations in medical imaging.

The dataset description table provides further insights into the characteristics of these datasets. Most of the publicly available datasets consist of WSIs stained with H&E, a widely used staining method in pathology. Many datasets also include immunohistochemistry-stained (IHC) images, which are crucial for studying tumor microenvironments, such as in LYSTO and BCIHC.

### *E. TRAINING PROCEDURES AND OPTIMIZATION STRATEGIES*

The training procedures and optimization strategies adopted in the reviewed studies are summarized in Table 18. Different types of optimization algorithms were employed, with Adam being the most commonly used optimizer, appearing in 43.59% of the studies. SGD was the second most utilized optimizer, used in 25.64% of the studies, followed by




(a) Encoder. (b) Combination Strategy. (c) Feature Fusion.

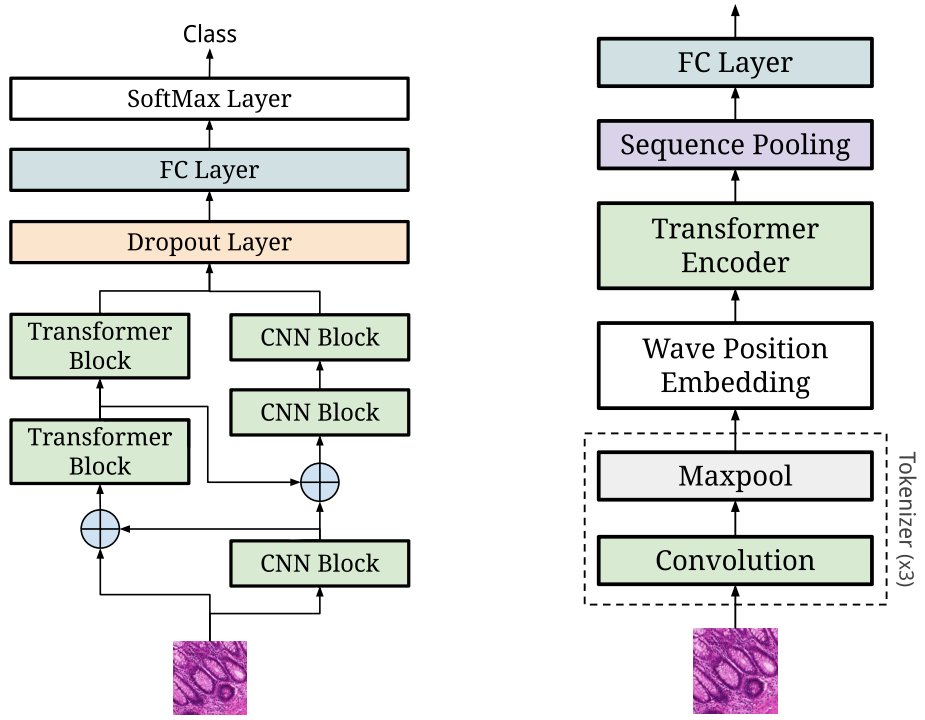
**FIGURE 12.** Distribution of architectural components in classification studies. (a) Encoder types show a predominance of hybrid architectures. (b) Combination strategies highlight the preference for sequential strategies, followed by parallel and integrated approaches. (c) Feature fusion techniques, where a significant portion of models do not employ feature fusion, while attention-based, simple, and adaptive fusion strategies are also utilized.

**TABLE 15.** Characteristics of the classification studies based on class architecture components.

| Component | Studies, n (%) | References |
|---|---|---|
| **Encoder** | | |
| Hybrid | 22 (100.0%) | [23], [36], [37], [103]–[121] |
| **Combination Strategy** | | |
| Sequential | 11 (50.0%) | [23], [36], [104], [108]–[110], [112], [118]–[121] |
| Parallel | 6 (27.27%) | [37], [103], [105], [107], [114], [115] |
| Integrated | 5 (22.73%) | [106], [111], [113], [116], [117] |
| **Feature Fusion** | | |
| No feature fusion | 12 (54.55%) | [23], [36], [104], [108]–[110], [112], [113], [118]–[121] |
| Attention-based | 5 (22.73%) | [37], [103], [106], [107], [117] |
| Simple | 3 (13.64%) | [111], [114], [115] |
| Adaptive | 2 (9.09%) | [105], [116] |

**TABLE 16.** Characteristics of the detection studies based on network architecture components.

| Component | Studies, n (%) | References |
|---|---|---|
| **Backbone** | | |
| Hybrid | 2 (100.0%) | [122], [123] |
| **Neck** | | |
| CNN | 1 (50.0%) | [122] |
| Hybrid | 1 (50.0%) | [123] |
| **Head** | | |
| CNN | 2 (100.0%) | [122], [123] |
| **Combination Strategy** | | |
| Integrated | 1 (50.0%) | [123] |
| Parallel | 1 (50.0%) | [122] |
| **Feature Fusion** | | |
| Attention-based | 2 (100.0%) | [122], [123] |




**(a) A parallel combination strategy adapted from Wang et al. [37].**

**(b) A sequential architecture where a CNN-based encoder is followed by a transformer, adapted from Ding et al. [109].**

**FIGURE 13.** Comparison of classification model architectures: (a) and (b) represent two different CNN–Transformer integration strategies for image classification.

**TABLE 17.** Frequency of dataset usage in the analyzed studies, categorized by task and corresponding references.

| Dataset | Usage | Task | References |
|---|---|---|---|
| GLAS-2015 | 9 | Segmentation | [25], [89], [91]–[93], [95], [97], [100], [102] |
| BreakHis | 9 | Classification | [23], [108]–[111], [115], [119]–[121] |
| MoNuSeg | 5 | Segmentation | [91], [95], [97], [99], [101] |
| Bowl-2018 | 4 | Segmentation | [93]–[95], [102] |
| Chaoyang | 3 | Classification | [113], [117], [118] |
| Laryngeal | 2 | Classification | [105], [107] |
| Camelyon | 2 | Classification | [36], [37] |
| LC25000 | 2 | Classification | [112], [114] |
| Other datasets | 28 | - | - |

AdamW at 15.38%. Other optimizers, such as RMSprop, Adagrad and Ranger, were seldom used.

Regarding learning rate scheduling, 58.97% of the studies did not explicitly report employing any scheduling strategy. Among those that did, cosine annealing was the most commonly adopted approach, facilitating gradual learning rate decay to improve convergence [91], [113], [117], [123]. Other techniques included cyclic and exponential decay schedules [23], [101], [109], [111], which adaptively adjust the learning rate based on training dynamics. These results indicate that although adaptive learning rate scheduling is recognized as beneficial, it remains underutilized or underreported in many studies.

Weight initialization methods were infrequently discussed, with 74.36% of the studies omitting details on this aspect. Among the few that provided such information, most leveraged pre-trained backbones, typically initialized on ImageNet, to enhance convergence and stability during training [36], [89], [92], [104], [105], [107]. The lack of reporting on initialization procedures highlights a gap in reproducibility standards, as weight initialization can significantly affect training dynamics and model performance.

Early stopping, a common regularization technique for preventing overfitting, was also sparsely reported. While 74.36% of the studies did not mention its use, only 25.64% explicitly documented this practice. When employed, early stopping was typically implemented with fixed patience values to

**TABLE 18.** Characteristics of the studies based on optimizer, learning rate scheduler, weight initialization, early stopping, and data augmentation.

| Feature | Studies, n (%) | References |
|---|---|---|
| **Optimizer Used** | | |
| Adam | 17 (43.59%) | [23], [37], [92]–[95], [97], [102], [103], [110]–[113], [115], [117], [120], [121] |
| SGD | 10 (25.64%) | [25], [89], [91], [96], [98]–[100], [104], [116], [122] |
| AdamW | 6 (15.38%) | [90], [101], [108], [109], [118], [123] |
| RMSprop | 2 (5.13%) | [105], [107] |
| Adagrad | 1 (2.56%) | [119] |
| Ranger optimizer | 1 (2.56%) | [36] |
| Not Mentioned | 2 (5.13%) | [106], [114] |
| **Learning Rate Scheduler** | | |
| Yes | 16 (41.03%) | [23], [89]–[92], [94], [101], [104], [107]–[109], [113], [117], [119], [120], [123] |
| Not Mentioned | 23 (58.97%) | [25], [36], [37], [93], [95]–[100], [102], [103], [105], [106], [110]–[112], [114]–[116], [118], [121], [122] |
| **Weight Initialization** | | |
| Yes | 10 (25.64%) | [23], [36], [89], [92], [104], [105], [107], [110], [111], [115] |
| Not Mentioned | 29 (74.36%) | [25], [37], [90], [91], [93]–[103], [106], [108], [109], [112]–[114], [116]–[123] |
| **Early Stopping** | | |
| Yes | 10 (25.64%) | [25], [36], [37], [92], [94], [103], [106], [114], [119], [123] |
| Not Mentioned | 29 (74.36%) | [23], [89]–[91], [93], [95]–[102], [104], [105], [107]–[113], [115]–[118], [120]–[122] |
| **Data Augmentation Techniques** | | |
| Yes | 25 (64.1%) | [23], [25], [36], [37], [89], [91], [92], [94], [95], [97], [101], [102], [104], [107]–[111], [113], [117], [118], [120]–[123] |
| Not Mentioned | 14 (35.9%) | [90], [93], [96], [98]–[100], [103], [105], [106], [112], [114]–[116], [119] |

terminate training once the validation loss plateaued [25], [36], [37], [92], [94], [106]. These findings suggest either a limited adoption of early stopping or a general lack of documentation regarding this strategy.

Data augmentation techniques were applied in 64.1% of the studies, underscoring their importance in improving model generalization. Commonly used strategies included geometric transformations, such as flipping, rotation, cropping, and scaling [37], [89], [91], [107], [109], [113], as well as intensity-based methods like brightness adjustment, color jittering, Gaussian noise, and elastic deformation [92], [101], [123]. However, 35.9% of the studies did not mention any data augmentation, which may limit model robustness, especially when working with small or imbalanced datasets.

### F. INTERPRETABILITY AND EXPLAINABILITY IN MODEL DECISIONS

Among the studies that incorporated interpretability and explainability strategies, a range of techniques were employed to elucidate model decision-making. Grad-CAM was the most frequently used method, providing visual heatmaps of regions influencing model predictions [23], [105], [106], [107], [113], [117], [120]. In addition, customized visualization techniques, such as GN-CAM for tumor localization [96] and rollout-based attention visualizations [107], were explored.

However, the analysis of interpretability and explainability methods in the reviewed studies (see Table 19) reveals that most works (64.10%) focus primarily on predictive performance without addressing the transparency of model decisions. Although 35.90% of the studies employed explainability techniques, interest in this area remains limited. This reflects a broader tendency to prioritize performance metrics over model interpretability, highlighting the need for more consistent integration of explainability strategies in histopathological AI pipelines, particularly for clinical decision support, where understanding model reasoning is essential.

### G. HARDWARE INFRASTRUCTURE AND COMPUTATIONAL TOOLS

The reviewed studies exhibit a strong preference for GPU-based training, with most utilizing dedicated hardware to support DL workloads. While some studies relied on single-GPU setups [23], [89], [94], [115], [122], others leveraged multi-GPU configurations or high-performance computing clusters to accelerate training times [36], [99], [107], [112], [121], [123]. Additionally, a subset of studies employed cloud-based services or institutional shared computing environments, underscoring the growing accessibility and scalability of computational infrastructure in histopathological AI research [110], [111], [118].

**TABLE 19.** Characteristics of the studies based on use of methods of model interpretability and explainability.

| Feature | Studies, n (%) | References |
|---|---|---|
| **Model Interpretability and Explainability** | | |
| Yes | 14 (35.9%) | [23], [36], [37], [89], [91], [94], [96], [102], [105]–[107], [113], [117], [120] |
| No | 25 (64.1%) | [25], [90], [92], [93], [95], [97]–[101], [103], [104], [108]–[112], [114]–[116], [118], [119], [121]–[123] |

**TABLE 20.** Characteristics of the studies based on used software libraries and source code availability.

| Feature | Studies, n (%) | References |
|---|---|---|
| **Software Libraries** | | |
| PyTorch | 26 (66.67%) | [25], [36], [37], [89]–[95], [97]–[99], [101]–[105], [107]–[110], [116], [119], [122], [123] |
| TensorFlow | 5 (12.82%) | [111], [112], [115], [118], [120] |
| Not Mentioned | 8 (20.51%) | [23], [96], [100], [106], [113], [114], [117], [121] |
| **Code Availability** | | |
| Yes | 8 (20.51%) | [94], [96], [97], [107], [108], [112], [120], [121] |
| No | 31 (79.49%) | [23], [25], [36], [37], [89]–[93], [95], [98]–[106], [109]–[111], [113]–[119], [122], [123] |

The analysis of computational tools used in the reviewed studies (see Table 20) shows a clear predominance of PyTorch, adopted in 66.67% of the cases. This likely reflects its flexibility, support for dynamic computation graphs, and strong adoption within the computer vision research community, factors that facilitate rapid prototyping of hybrid architectures. In contrast, TensorFlow appeared in 12.82% of the studies, indicating moderate use. Notably, 20.51% of the papers did not report the framework employed, which limits reproducibility and transparency.

Despite the increasing emphasis on open science, 79.49% of the studies did not provide access to their implementation, thereby limiting transparency and reproducibility. Only 20.51% made their source code publicly available. Encouraging broader adoption of open-source practices, including code sharing, pretrained models, and experiment logs, could significantly improve reproducibility, facilitate benchmarking, and foster collaboration across research groups.

### *H. PERFORMANCE EVALUATION*

The performance evaluation of the models was conducted using different strategies, as illustrated in Fig. 14 and detailed in Table 21.

Among the reviewed studies, holdout validation was the most commonly used strategy (84.62%), typically involving training/testing splits such as 70/30. Although straightforward to implement, this approach may yield optimistic performance estimates due to potential overfitting on small datasets [89], [109], [120]. In contrast, cross-validation, used in only 28.21% of studies, offers a more robust assessment of generalizability by averaging performance over multiple folds [104], [106], [107]. External validation, implemented in 25.64% of the studies, represents the gold standard for evaluating model performance on independent cohorts [36], [98], though its limited adoption highlights barriers in dataset availability and harmonization.

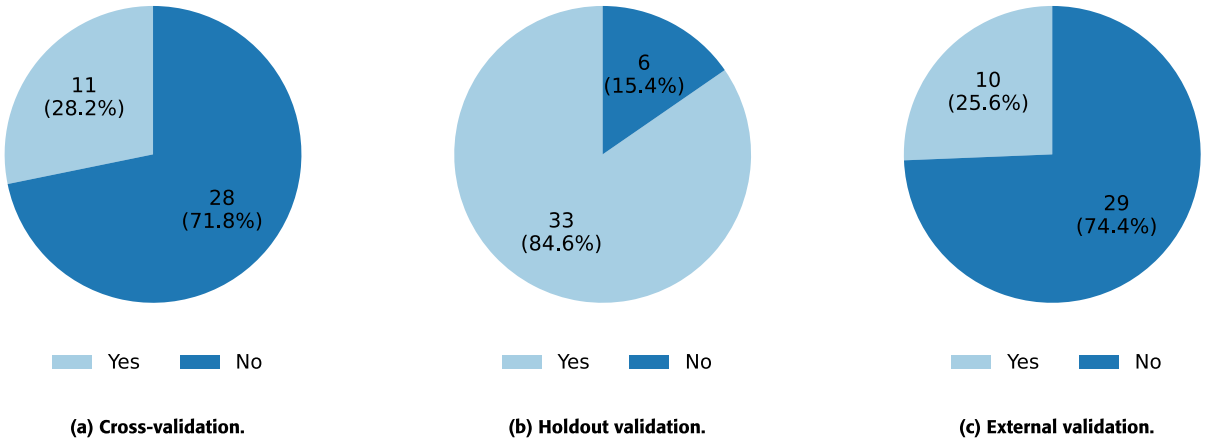
Fig. 14 graphically highlights the distribution of evaluation strategies among the studies, while Table 21 provides detailed characteristics and references for each approach. These analyses reflect the diversity of methods used to ensure the robustness and applicability of models in different application contexts. Moreover, the limited use of cross- and external validation raises concerns about the real-world generalizability of many models. Future studies should prioritize these strategies to ensure robustness across institutional, demographic, and staining protocol variations.

In addition to validation strategies, the choice of performance metrics varied significantly by task. For segmentation models (Table 22), there was a strong consensus on using overlap-based metrics. The Dice coefficient (93.33%) and Intersection over Union (IoU) (80.0%) were overwhelmingly preferred. Other metrics, such as Sensitivity (26.67%), Precision (26.67%), and Accuracy (20.0%), were less frequent, while metrics assessing boundary quality, like the Hausdorff Distance (20.0%), were rare.

For classification tasks (Table 23), Accuracy was the most dominant metric, reported in 95.45% of studies. This was commonly supplemented by F1-score (81.82%), Sensitivity (81.82%), and Precision (72.73%) to provide a more robust evaluation, particularly in the context of potentially imbalanced data. The area under the curve (AUC)

**TABLE 21.** Characteristics of the studies based on evaluation strategies.

| Feature | Studies, n (%) | References |
|---|---|---|
| **Cross-Validation** | | |
| Yes | 11 (28.21%) | [23], [37], [95], [103], [104], [106], [107], [114], [118], [119], [121] |
| No | 28 (71.79%) | [25], [36], [89]–[94], [96]–[102], [105], [108]–[113], [115]–[117], [120], [122], [123] |
| **Holdout Validation** | | |
| Yes | 33 (84.62%) | [25], [36], [37], [89]–[94], [96]–[103], [105]–[114], [116]–[118], [120], [122], [123] |
| No | 6 (15.38%) | [23], [95], [104], [115], [119], [121] |
| **External Validation** | | |
| Yes | 10 (25.64%) | [25], [36], [96], [98], [104], [105], [107], [108], [110], [122] |
| No | 29 (74.36%) | [23], [37], [89]–[95], [97], [99]–[103], [106], [109], [111]–[121], [123] |




(a) Cross-validation. (b) Holdout validation. (c) External validation.

**FIGURE 14.** Distribution of evaluation strategies used in the reviewed studies.

was also adopted by a majority of studies (54.55%). For the detection task (Table 24), F1-score and Sensitivity were used by all reviewed studies.

A significant finding of this review, summarized in Table 25, is the widespread underreporting of computational and statistical analyses. Only 23.08% of the studies reported computational costs (e.g., MACs, FLOPs), and only 35.9% disclosed the total number of model parameters. This lack of transparency hinders the assessment of model efficiency and reproducibility. Furthermore, an overwhelming majority of studies (94.87%) did not employ any statistical tests to validate the significance of their results.

### *I. STUDY AND MODEL LIMITATIONS*

This section outlines key limitations observed across the reviewed studies, distinguishing between challenges related to study design and data, and those inherent to model architectures and performance.

#### 1) STUDY LIMITATIONS

The reviewed studies exhibited several recurrent limitations, particularly concerning dataset availability and quality, model generalizability, and annotation practices.

A predominant issue relates to dataset constraints. Many studies relied on small, imbalanced, or highly domain-specific datasets, limiting the ability of models to generalize beyond their original training contexts [25], [36], [89], [91], [92]. Publicly available histopathology datasets, while essential for benchmarking, often lack intra-dataset diversity in staining protocols, disease subtypes, and acquisition equipment. Across different datasets, another key limitation is the overrepresentation of a few cancer types—such as breast [23], [36], [106], [107], [121], [122] and colon [108], [112], [113]—while others remain underrepresented. These factors significantly affect model robustness and clinical applicability.

Annotation-related challenges also emerged as a critical limitation. Manual annotations, typically performed by pathologists, are labor-intensive and time-consuming, constraining the size and quality of labeled data [104], [106]. In some cases, annotations focused on coarse or global labels, which limited the granularity of model supervision [104], [123]. Moreover, annotation errors — such as the selection of non-tumor regions or the generation of false-positive pseudo labels — can hinder reproducibility and model transferability [36].

**TABLE 22.** Usage of evaluation metrics in segmentation studies.

| Metric | Studies, n (%) | References |
|---|---|---|
| **Dice** | | |
| Yes | 14 (93.33%) | [25], [89]–[100], [102] |
| No | 1 (6.67%) | [101] |
| **IoU** | | |
| Yes | 12 (80.0%) | [25], [89]–[91], [93]–[99], [102] |
| No | 3 (20.0%) | [92], [100], [101] |
| **Accuracy** | | |
| Yes | 3 (20.0%) | [96], [97], [102] |
| No | 12 (80.0%) | [25], [89]–[95], [98]–[101] |
| **Sensitivity** | | |
| Yes | 4 (26.67%) | [25], [96], [97], [102] |
| No | 11 (73.33%) | [89]–[95], [98]–[101] |
| **Specificity** | | |
| Yes | 2 (13.33%) | [93], [96] |
| No | 13 (86.67%) | [25], [89]–[92], [94], [95], [97]–[102] |
| **Precision** | | |
| Yes | 4 (26.67%) | [25], [93], [96], [97] |
| No | 11 (73.33%) | [89]–[92], [94], [95], [98]–[102] |
| **AUC** | | |
| Yes | 1 (6.67%) | [25] |
| No | 14 (93.33%) | [89]–[102] |
| **Hausdorff Distance (HD)** | | |
| Yes | 3 (20.0%) | [25], [95], [98] |
| No | 12 (80.0%) | [89]–[94], [96], [97], [99]–[102] |
| **Panoptic Quality (PQ)** | | |
| Yes | 1 (6.67%) | [101] |
| No | 14 (93.33%) | [25], [89]–[100], [102] |
| **Mean Average Error (MAE)** | | |
| Yes | 1 (6.67%) | [90] |
| No | 14 (93.33%) | [25], [89], [91]–[102] |
| **Weighted F-measure** | | |
| Yes | 1 (6.67%) | [90] |
| No | 14 (93.33%) | [25], [89], [91]–[102] |
| **S-measure** | | |
| Yes | 1 (6.67%) | [90] |
| No | 14 (93.33%) | [25], [89], [91]–[102] |
| **E-measure** | | |
| Yes | 1 (6.67%) | [90] |
| No | 14 (93.33%) | [25], [89], [91]–[102] |

Generalization challenges were frequently highlighted. Several models showed strong performance on internal validation sets but struggled when applied to external or independent datasets [122]. This suggests overfitting to specific dataset characteristics, such as staining variations, image resolution, or institutional biases. Furthermore, a lack of external validation in many studies undermines the confidence in the real-world applicability of these models.

### 2) MODEL LIMITATIONS

From a modeling perspective, key limitations include high computational demands, dependency on large-scale annotated data, and difficulties in capturing complex morphological features.

Transformer-based models, particularly in hybrid CNN-Transformer architectures, often require substantial computational resources for both training and inference [89],

**TABLE 23.** Usage of evaluation metrics in classification studies.

| Metric | Studies, n (%) | References |
|---|---|---|
| **Accuracy** | | |
| Yes | 21 (95.45%) | [23], [36], [37], [103], [105]–[121] |
| No | 1 (4.55%) | [104] |
| **F1-score** | | |
| Yes | 18 (81.82%) | [23], [36], [103]–[107], [109]–[118], [121] |
| No | 4 (18.18%) | [37], [108], [119], [120] |
| **Precision** | | |
| Yes | 16 (72.73%) | [23], [36], [104]–[106], [109]–[118], [121] |
| No | 6 (27.27%) | [37], [103], [107], [108], [119], [120] |
| **Sensitivity** | | |
| Yes | 18 (81.82%) | [23], [36], [103]–[107], [109]–[118], [121] |
| No | 4 (18.18%) | [37], [108], [119], [120] |
| **AUC** | | |
| Yes | 12 (54.55%) | [36], [37], [103], [106], [107], [109], [112], [113], [117]–[119], [121] |
| No | 10 (45.45%) | [23], [104], [105], [108], [110], [111], [114]–[116], [120] |
| **Specificity** | | |
| Yes | 5 (22.73%) | [103], [107], [112], [118], [121] |
| No | 17 (77.27%) | [23], [36], [37], [104]–[106], [108]–[111], [113]–[117], [119], [120] |
| **Precision-Recall Curve** | | |
| Yes | 1 (4.55%) | [112] |
| No | 21 (95.45%) | [23], [36], [37], [103]–[111], [113]–[121] |
| **Matthews Correlation Coefficient (MCC)** | | |
| Yes | 3 (13.64%) | [23], [111], [121] |
| No | 19 (86.36%) | [36], [37], [103]–[110], [112]–[120] |
| **Cohen's kappa coefficient** | | |
| Yes | 2 (9.09%) | [23], [121] |
| No | 20 (90.91%) | [36], [37], [103]–[120] |

**TABLE 24.** Usage of evaluation metrics in detection studies.

| Metric | Studies, n (%) | References |
|---|---|---|
| **F1-score** | | |
| Yes | 2 (100.0%) | [122], [123] |
| No | 0 (0.0%) | |
| **Precision** | | |
| Yes | 1 (50.0%) | [123] |
| No | 1 (50.0%) | [122] |
| **Sensitivity** | | |
| Yes | 2 (100.0%) | [122], [123] |
| No | 0 (0.0%) | |

[91]. This complexity is further exacerbated when dealing with high-resolution WSIs, which significantly increase memory and processing requirements [37]. Although some studies employed pre-trained backbones to mitigate training costs, fine-tuning still incurs notable overhead, limiting accessibility in resource-constrained settings.

Another limitation is the sensitivity of ViT-based models to dataset size and image quality. Several studies reported

**TABLE 25.** Reporting of computational and statistical aspects across classification, segmentation, and detection studies.

| Reported item | Studies, n (%) | References |
|---|---|---|
| **Computational cost** | | |
| Yes | 9 (23.08%) | [90], [92], [96], [101], [102], [109], [114], [118], [124] |
| No | 30 (76.92%) | [23], [25], [36], [37], [91], [93]–[95], [97]–[100], [103]–[108], [110]–[113], [115]–[117], [119]–[123] |
| **Parameters** | | |
| Yes | 14 (35.9%) | [92], [93], [96], [98], [100]–[103], [108], [109], [114], [118], [119], [124] |
| No | 25 (64.1%) | [23], [25], [36], [37], [90], [91], [94], [95], [97], [99], [104]–[107], [110]–[113], [115]–[117], [120]–[123] |
| **Statistical test** | | |
| Yes | 2 (5.13%) | [90], [92] |
| No | 37 (94.87%) | [23], [25], [36], [37], [91], [93]–[124] |

performance degradation when training with limited or imbalanced data, highlighting a persistent dependency on large, diverse datasets [113]. In addition, Transformers often struggle with subtle textural details and local morphological variations, which are crucial in histopathology, particularly when the patch size exceeds the scale of discriminative structures [92], [108].

Overfitting also remains a challenge, especially when training complex hybrid architectures on small datasets. Despite the potential of attention mechanisms to enhance feature representation, they can inadvertently increase model capacity, amplifying the risk of memorization rather than generalization [94], [107].

Lastly, relatively few studies systematically evaluated the behavior of models under varying clinical conditions, such as heterogeneous tissue types, rare tumor subtypes, or different acquisition devices. This gap underscores the need for more rigorous stress-testing protocols to assess robustness and reliability prior to clinical deployment.

## V. DISCUSSION

This section discusses the findings of this scoping review in light of the research questions (RQs) presented in Section I. The discussion is structured according to the main research questions, integrating insights from the included studies and contextualizing them with relevant literature.

### A. HYBRID CNN-TRANSFORMER ARCHITECTURES IN HISTOPATHOLOGY IMAGE ANALYSIS (RQ1)

The analyzed studies reveal a diverse set of architectural strategies for integrating CNNs and Transformers, broadly categorized as sequential, parallel, and integrated combinations. These designs leverage the locality-aware feature extraction of CNNs and the global context modeling capabilities of Transformers.

Across all tasks (Table 13), classification emerged as the most frequent application, followed by segmentation and, to a lesser extent, detection. Fully supervised learning predominated, with fewer studies adopting weakly supervised, semi-supervised, or self-supervised approaches. A substantial portion of the models were explicitly designed for histopathology, reflecting a growing emphasis on domain-specific architectural adaptations.

Classification architectures (Table 15) consistently employed hybrid encoders. Sequential combination strategies were the most common, typically using a CNN to extract local features followed by a Transformer for capturing long-range dependencies. Parallel and integrated strategies were also present but less frequent. Many classification models did not include explicit feature fusion modules, reflecting the linear flow of information in sequential pipelines. Among those that did employ feature fusion, attention-based approaches were the most common, followed by simple and adaptive fusion.

Segmentation architectures (Table 14) displayed greater diversity in component design. Hybrid encoders and bottlenecks were prevalent, although CNN-based decoders were more common. Skip connections often relied on custom modules to better integrate multi-scale features. Sequential strategies remained the most frequent, but parallel and integrated designs were also well represented, reflecting the multi-stream processing requirements of dense prediction tasks. While some segmentation models did not employ feature fusion, others favored simple or attention-based fusion, with adaptive fusion appearing only occasionally.

Detection architectures (Table 16) were based exclusively on hybrid backbones combined with CNN-based heads. Neck designs varied between CNN and hybrid modules. Combination strategies included both integrated and parallel approaches. All detection models incorporated attention-based feature fusion, suggesting its perceived importance in object localization tasks.

When comparing tasks, sequential strategies tend to dominate in classification, likely due to their simpler design and effective integration of local and global features. Segmentation exhibits a more balanced distribution of sequential, parallel, and integrated approaches, often combined with

custom skip connections and fusion modules to handle spatial detail and multi-scale context. Detection, although less represented, consistently employs hybrid backbones and attention-based fusion, indicating a prioritization of precise feature alignment for object localization. These patterns highlight that the choice of integration strategy and architectural components is strongly driven by the specific requirements of each histopathology task.

### B. DATASETS UTILIZED IN HYBRID CNN-TRANSFORMER MODELS (RQ2)

The reviewed studies utilized various datasets for training and evaluation, with a predominance of publicly available histopathology datasets such as GLAS-2015, BreakHis, MoNuSeg, and Camelyon [92]. These datasets provide essential benchmarks for segmentation, classification, and detection tasks, yet they exhibit certain limitations.

One key challenge is dataset size, as many histopathology datasets contain limited samples, potentially affecting the generalization of DL models. To mitigate this, data augmentation techniques, such as flipping, rotation, and color jittering, were widely employed [96], [120]. However, the lack of standardization in augmentation methods across studies complicates direct comparisons of model performance.

Moreover, dataset annotation quality varies significantly, impacting the robustness of hybrid models [11]. Semi-supervised and weakly supervised learning approaches have been explored to address annotation constraints [37], [103], but their adoption remains limited in histopathology. Future research should prioritize strategies for improving dataset diversity and annotation reliability.

### C. EVALUATION METRICS AND PERFORMANCE COMPARISONS (RQ3)

The analysis of evaluation metrics (RQ3) confirms that performance criteria are strictly task-dependent, reflecting established standards within each domain, as detailed in Tables 22–24. For segmentation, a strong consensus was observed on overlap-based metrics, with the Dice coefficient (93.33%) and Intersection over Union (IoU) (80.0%) being the predominant choices for quantifying spatial evaluation [89], [91]. In classification studies, Accuracy (95.45%) was almost universally reported. However, acknowledging the challenges of often-imbalanced medical datasets, most studies also adopted metrics such as F1-score (81.82%), Sensitivity (81.82%), and Precision (72.73%) to provide a more robust and comprehensive assessment of model performance [36], [110].

Beyond the choice of metrics, a critical gap was identified in the reporting of model complexity and statistical validation (see Table 25). There is a significant lack of transparency regarding computational efficiency; only 23.08% of studies reported computational costs, and only 35.9% disclosed the total number of parameters. This omission hinders the assessment of model efficiency and reproducibility, which are crucial factors for practical clinical deployment. Furthermore, the absence of statistical significance testing was almost universal, with only 5.13% of studies employing such tests. This omission makes it difficult to ascertain whether the reported performance improvements are statistically meaningful, representing a critical gap in the reviewed literature.

A notable observation is the inconsistent use of external validation datasets. Many studies relied solely on internal validation, limiting the generalization of their findings [113], [123]. The adoption of cross-validation and external benchmarking datasets should be encouraged to improve model reliability.

Additionally, interpretability and explainability remain underexplored in many studies. While methods such as Grad-CAM [106], [120] were used to provide visual explanations, only a minority of studies reported utilizing such techniques. The integration of explainability frameworks into hybrid CNN-Transformer models is essential for clinical adoption and trustworthiness [97].

### D. CHALLENGES AND FUTURE DIRECTIONS (RQ4 AND RQ5)

Despite the promising results of hybrid CNN-Transformer models, several significant challenges persist, particularly concerning computational cost and architectural design, which are critical barriers to clinical applications. A major limitation is the computational expense associated with Transformer-based architectures [105]. While pre-training can mitigate this, the high memory and processing demands for fine-tuning and inference remain a primary obstacle [103]. To address this, future research must focus on model efficiency through several complementary strategies. One primary approach is the adoption of inherently efficient architectures, such as lightweight Transformer variants like Swin Transformers [117] or deformable Transformers [11].

Another crucial possibility is the application of model compression techniques, particularly pruning. This involves more than simply removing parameters; an important distinction exists between removing individual weights (weight pruning) and entire structural components (channel pruning) [124], the latter being more suitable for hybrid models. A significant challenge, however, is the high computational cost. Traditionally, pruning is performed after training, which mandates an expensive retraining or fine-tuning phase to regain lost accuracy [124], [125]. This step also introduces complex hyperparameters, such as determining the optimal pruning rate. A promising research direction, therefore, is the investigation of methods that prune before or during training, which aim to find efficient models without this costly retraining cycle [125].

While pruning targets the model's structural redundancy, it can be combined with quantization to maximize efficiency. This complementary technique reduces the numerical precision of model weights and activations, typically from 32-bit floating point (FP32) to 8-bit integer (INT8) [126]. The

primary benefit is a significant reduction in the model's memory footprint and the potential for faster inference, especially on hardware with specialized support for low-precision arithmetic. For hybrid models, this can be applied to both the CNN and Transformer components. However, as highlighted by reviews like [127], this presents a key challenge. The simplest method, post-training quantization (PTQ), is computationally cheap but often results in a significant accuracy drop. This is because the self-attention mechanisms in Transformers are particularly sensitive to reduced precision, due to factors like unbalanced data distributions after the softmax function [127]. Therefore, to maintain high accuracy, a more robust method is often necessary: quantization-aware training (QAT). QAT simulates the lower precision during the fine-tuning process, allowing the model to adapt, but at the cost of a full retraining cycle [126], [127].

Finally, knowledge distillation is a compression technique where a large ''teacher'' model trains a smaller ''student'' model [128]. This is highly relevant for hybrid architectures, which are often used as powerful, but complex, ''teacher'' models. This process can be used to train a smaller, efficient hybrid student, or, as demonstrated by Song et al. [129], an ensemble ''teacher'' combining a CNN and a ViT can be used to distill knowledge into a single, lightweight CNN ''student''. This approach is highly effective because the hybrid teacher combines the CNN's strong inductive bias for local features with the ViT's ability to model global, long-range dependencies. The student model (even if it is just a simple CNN) is then trained to mimic this superior, combined output, resulting in a model that is both compact and highly accurate [127], [129].

Beyond efficiency, a more fundamental challenge and a key finding of this review, is the lack of consensus on optimal hybrid architectural design. As our analysis shows, the literature is fragmented, making direct comparisons difficult [101]. A critical analysis reveals several unresolved questions regarding the design of these hybrid systems. For example, there is no clear evidence on the superiority of one combination strategy. For classification studies (Table 15), the field is almost evenly split between sequential (50.0%) and non-sequential designs (27.27% parallel and 22.73% integrated). A similar fragmentation is seen in segmentation (Table 14), with sequential (46.67%) and parallel (33.33%) approaches both being prevalent.

This division reflects a fundamental design trade-off. Sequential models are conceptually straightforward, leveraging CNNs for robust local feature extraction and subsequently using Transformers for global context modeling. Their primary weakness, however, is the risk of an information bottleneck, where the Transformer's performance is entirely dependent on the quality of the features provided by the initial CNN encoder. On the contrary, parallel architectures attempt to mitigate this by allowing both modules to learn complementary representations (e.g., local and global) independently from the input. Their main limitations, however, include a significant increase in parameter count and computational cost, the risk of co-training instability between branches and increased fusion complexity during feature integration.

These documented design inconsistencies across all tasks underscore the urgent need for standardized evaluation protocols. Without common benchmarks and datasets, it is nearly impossible to fairly compare the merits of different hybridization strategies [101]. Looking forward, promising directions include the integration of multi-modal data, such as combining histopathology with genomic or radiological information [119], and the exploration of multi-task learning [118]. Hybrid models may be particularly well-suited for such tasks, potentially allowing the CNN branch to focus on segmentation while the Transformer branch handles classification, with their outputs fused for a comprehensive diagnosis.

## VI. CONCLUSION

This scoping review offers a comprehensive synthesis of hybrid CNN–Transformer architectures in histopathological image analysis. Through the analysis of 39 peer-reviewed studies, we addressed three central objectives: (i) to categorize architectural configurations, (ii) to analyze their deployment across classification, segmentation, and detection tasks, and (iii) to identify methodological gaps and limitations that hinder their advancement toward clinical use.

In response to the first objective, we proposed a task-specific taxonomy that classifies hybrid models according to their primary applications. This taxonomy revealed that most studies adopt integrated or sequential architectures, leveraging the complementary strengths of CNNs and Transformers. These configurations are particularly prevalent in segmentation tasks, where performance gains are more consistently observed. However, the architectural diversity across studies remains limited, and strategies such as cross-guided learning and multi-path fusion are underexplored.

Regarding the second objective, our analysis showed that hybrid models have been successfully applied to the three core tasks—segmentation, classification, and detection—with classification receiving the greatest attention. Despite promising results, performance assessments often lack external validation or comparative baselines, and many studies rely on small, homogeneous datasets, limiting generalizability.

Addressing the third objective, we identified several critical methodological limitations. Most studies employ fully supervised learning with holdout validation, while cross-validation and external testing are less explored. Data augmentation is common but is applied and reported inconsistently. Interpretability remains an overlooked aspect: only 35.9% of the studies employed explainability techniques, usually limited to Grad-CAM visualizations. Moreover, reproducibility is compromised by the scarcity of publicly available code and incomplete reporting of implementation details.

In this context, this review highlights the growing interest in hybrid CNN-Transformer models for histopathological image analysis and the need for more rigorous, standardized, and transparent research practices. However, this review also has limitations. Our search was restricted to peer-reviewed journal articles in English from four databases, potentially excluding emergent models in preprints or non-English publications. Consistent with scoping review methodology, we did not perform a formal quality appraisal of the included studies, and new articles may have been published after our search cutoff date.

Future work should prioritize the development of lightweight, interpretable architectures, the use of diverse and well-annotated datasets, and adherence to standardized evaluation protocols. The proposed taxonomy and the critical mapping provided in this review establish a foundation to guide future investigations and facilitate the transition of hybrid models from experimental design to clinical utility.

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**DOMINGOS LUCAS LATORRE DE OLIVEIRA** received the B.Tech. degree in information technology (business management) from the Faculdade de Tecnologia de Praia Grande (FATEC–Praia Grande), Brazil, in 2006, and the M.Sc. degree in information engineering from the Federal University of ABC (UFABC), Brazil, in 2013. He is currently pursuing the Ph.D. degree in computer science with the Federal University of Uberlândia (UFU), Brazil, under the supervision of Prof. Marcelo Zanchetta do Nascimento.

Since 2010, he has been a Professor with the Federal Institute of Education, Science and Technology of São Paulo (IFSP), São Paulo, Brazil. His research interests include digital image processing and computer vision, with emphasis on medical and histopathological images, segmentation, feature extraction/selection, and machine learning for computer-aided diagnosis (CAD). He has authored and co-authored academic works in the aforementioned areas and contributes to interdisciplinary projects in biomedical imaging.

**THAÍNA APARECIDA AZEVEDO TOSTA** received the B.S. and M.S. degrees in computer science from the Federal University of Uberlândia (UFU), Brazil, and the Ph.D. degree in information engineering from the Federal University of ABC (UFABC), Brazil.

She is currently with the Institute of Science and Technology (ICT), Federal University of São Paulo (UNIFESP), São José dos Campos, Brazil. Her work spans medical image processing and analysis, including normalization, segmentation, feature extraction and selection, and classification for computer-aided diagnosis (CAD) in histological (H&E), hematological, and MR images. Her research interests include artificial intelligence for medical imaging, machine learning, and pattern recognition.

**LEANDRO ALVES NEVES** received the B.Tech. degree in data processing from the Centro Universitário de Rio Preto (UNIRP), Brazil, in 1998, and the M.Sc. and Ph.D. degrees in electrical engineering from the University of São Paulo (USP), Brazil, in 2001 and 2005, respectively.

He also completed postdoctoral research focusing on the computational analysis of myocardial biopsy images. He is currently an Associate Professor of applied computer science with São Paulo State University (UNESP), Brazil. His research interests include digital image processing, machine learning, and computational intelligence, with applications in medical imaging, biological systems, and complex data modeling.

Dr. Neves actively contributes as a reviewer and an Advisory Committee Member of leading Brazilian research funding agencies, such as FAPESP and CNPq, and serves on the editorial and review boards for international journals in machine learning, pattern recognition, and medical imaging.

**MARCELO ZANCHETTA DO NASCIMENTO** received the B.Tech. degree in data processing from the Centro Universitário de Rio Preto (UNIRP), Brazil, in 1998, the M.Sc. degree in electrical engineering from the University of São Paulo (USP), Brazil, in 2002, and the Ph.D. degree in electrical engineering from USP, São Carlos Campus, Brazil, in 2005.

He is currently a Full Professor with the Faculty of Computer Science (FACOM), Federal University of Uberlândia (UFU), Brazil. His research interests include biomedical engineering and medical imaging, with emphasis on digital image processing for radiographic and histopathological images, X-ray imaging physics (e.g., heel effect), simulation, and algorithm development. He has extensive service as a reviewer and editorial board member across numerous journals and agencies.

Dr. Nascimento serves as a reviewer and a Committee Member for research funding agencies, such as FAPESP, CNPq, and CAPES, and for a wide range of journals spanning computer vision, biomedical engineering, and medical imaging.

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